

X-ray Diffraction

Discovery of X-ray

W. K. Rontgen Physical Institute, University of Wurzburg



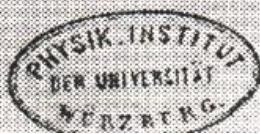
• 8 Nov. 1895



Hand mit Röntgen S. 2. 4. 14.



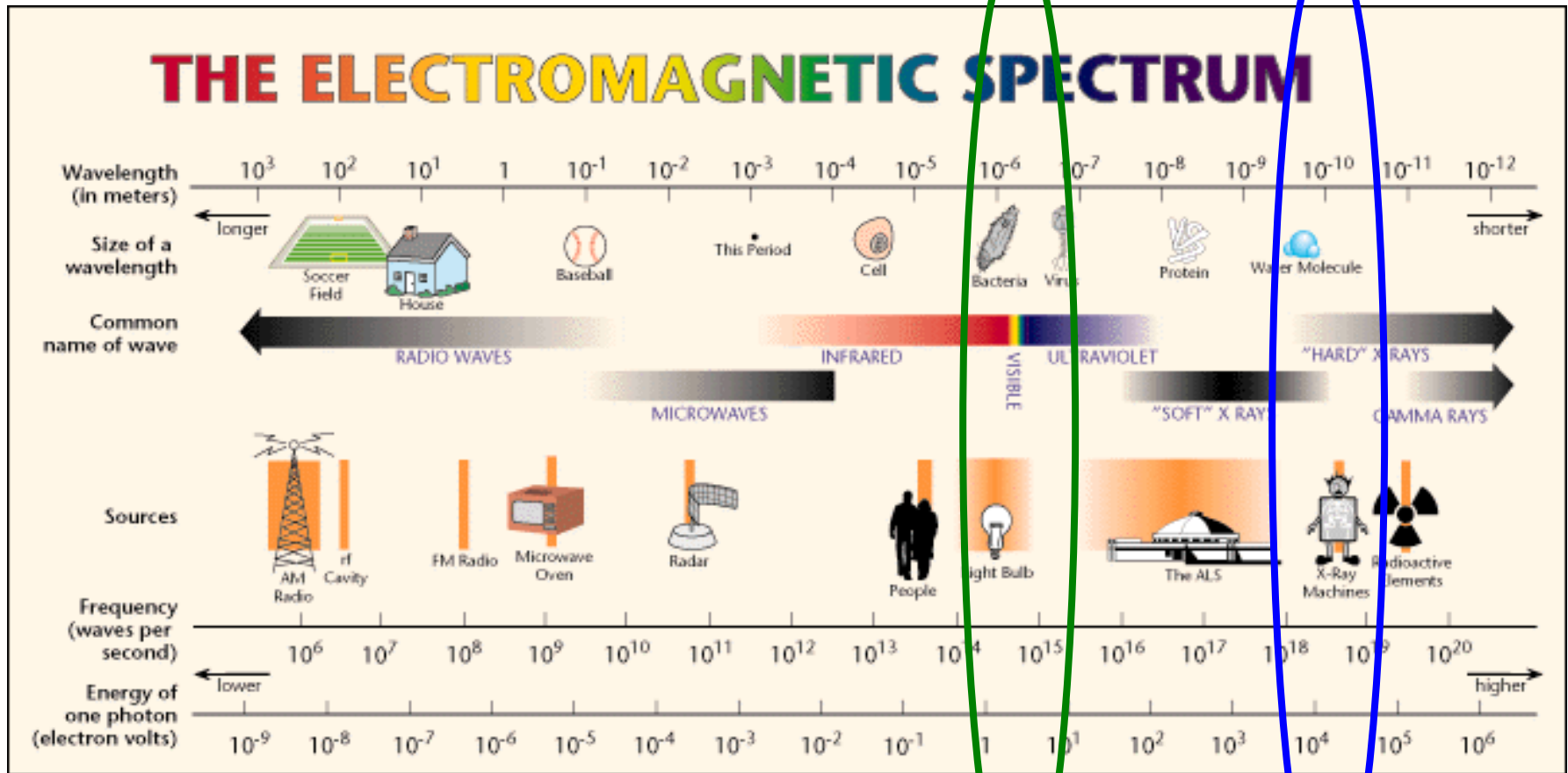
*Eigentum von Prof. Lehmann
Freiburg i. B.*



- 22 Dec.
hand of Frau Rontgen
- 28 Dec. First paper
“Eine Neue Art von Strahlen”
“一种新的射线----初步报告”
维茨堡物理学医学会会刊
- 5 Jan. 1896
“X-ray discovered by Rontgen”
维也纳新闻报
- 23 Jan
Alber von Kolliker suggested
Rontgen ray
1901 First Nobel prize for Physics
- 100 years**
tens Nobel Prize winners

Diffraction gratings

Lattice spacings
Crystal Structures



$$\lambda = \frac{c}{f}$$

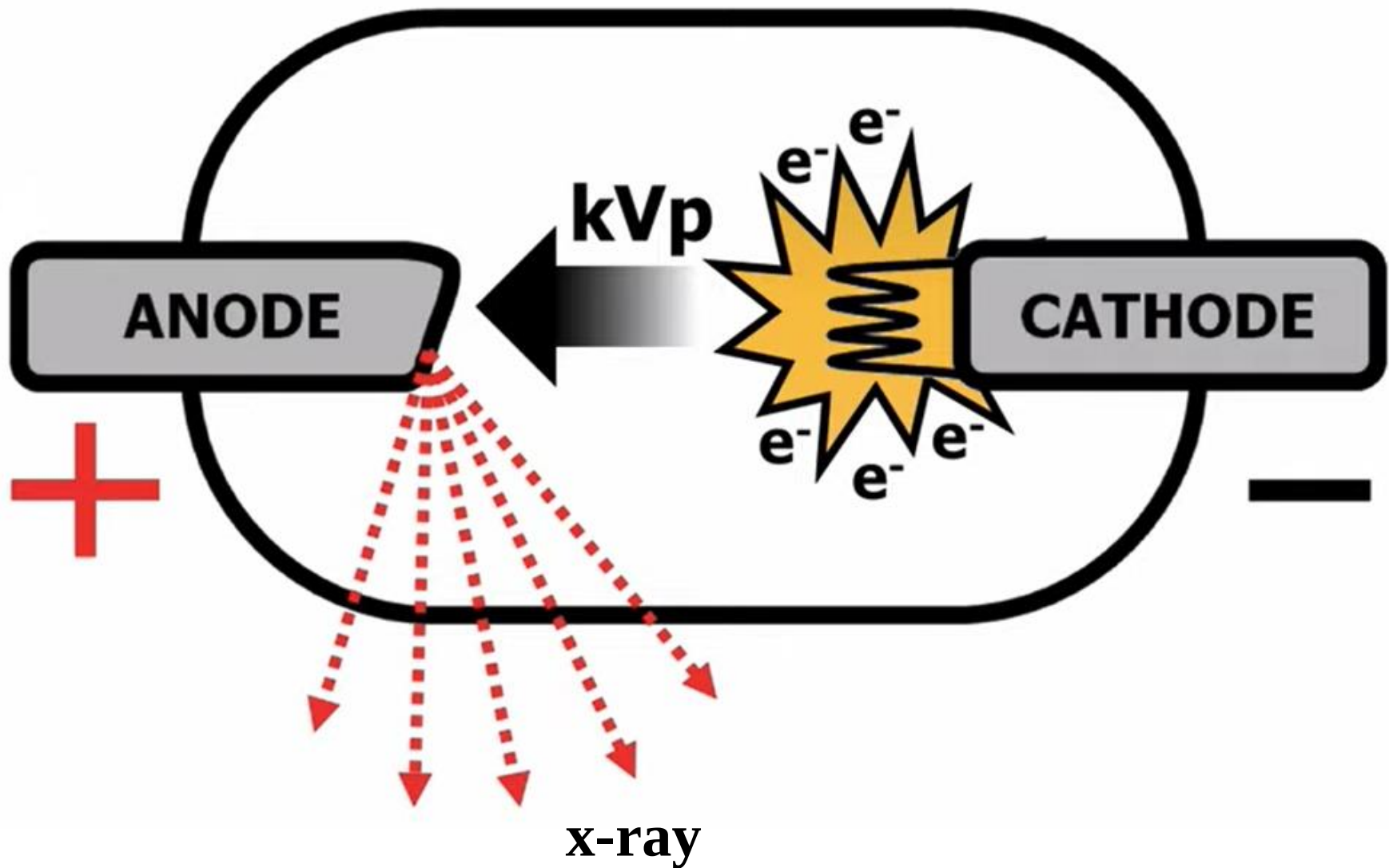
$$E = hf = \frac{hc}{\lambda}$$

Advanced Light Source

Optical diffraction

X-ray diffraction

The x-ray production



The basic method of crystal diffraction

1. X-ray diffraction

X-ray is an electromagnetic wave generated by electrons accelerated by a high voltage striking the "target" material

$$h\gamma_{\max} = eU \quad h\frac{c}{\lambda_{\min}} = eU$$

$$\lambda_{\min} = \frac{hc}{eU} \approx \frac{1.2 \times 10^3}{U} \text{ (nm)}$$

$$h \approx 6.62 \times 10^{-34} \text{ J} \cdot \text{s} \quad c = 3 \times 10^8 \text{ m/s} \quad e \approx 1.6 \times 10^{-19} \text{ C}$$

$$U = 10^4 \text{ V}, \quad \lambda \sim 0.1 \text{ nm}$$

In practice, normally 40 kV is applied

The basic method of crystal diffraction

2. Electron diffraction

$$\lambda = \frac{h}{P}, \quad \frac{P^2}{2m} = eU, \quad P = \sqrt{2meU}, \quad \lambda = \frac{h}{\sqrt{2meU}}$$

$$h \approx 6.62 \times 10^{-34} \text{ J} \cdot \text{s} \quad e \approx 1.6 \times 10^{-19} \text{ C} \quad m \approx 9.1 \times 10^{-31} \text{ kg}$$

$$\lambda \approx \sqrt{\frac{1.5}{U}} \text{ (nm)} \quad U = 150 \text{ V}, \lambda \sim 0.1 \text{ nm}$$

Electron waves are scattered by electrons and nuclei, with strong scattering and weak transmission

The basic method of crystal diffraction

3. Neutron diffraction

$$m_{\text{中}} \approx 2000m_{\text{电}} \quad \lambda \approx \sqrt{\frac{1.5}{2000U}} \approx \sqrt{\frac{7.5}{U}} \times 10^{-2} \text{ nm}$$

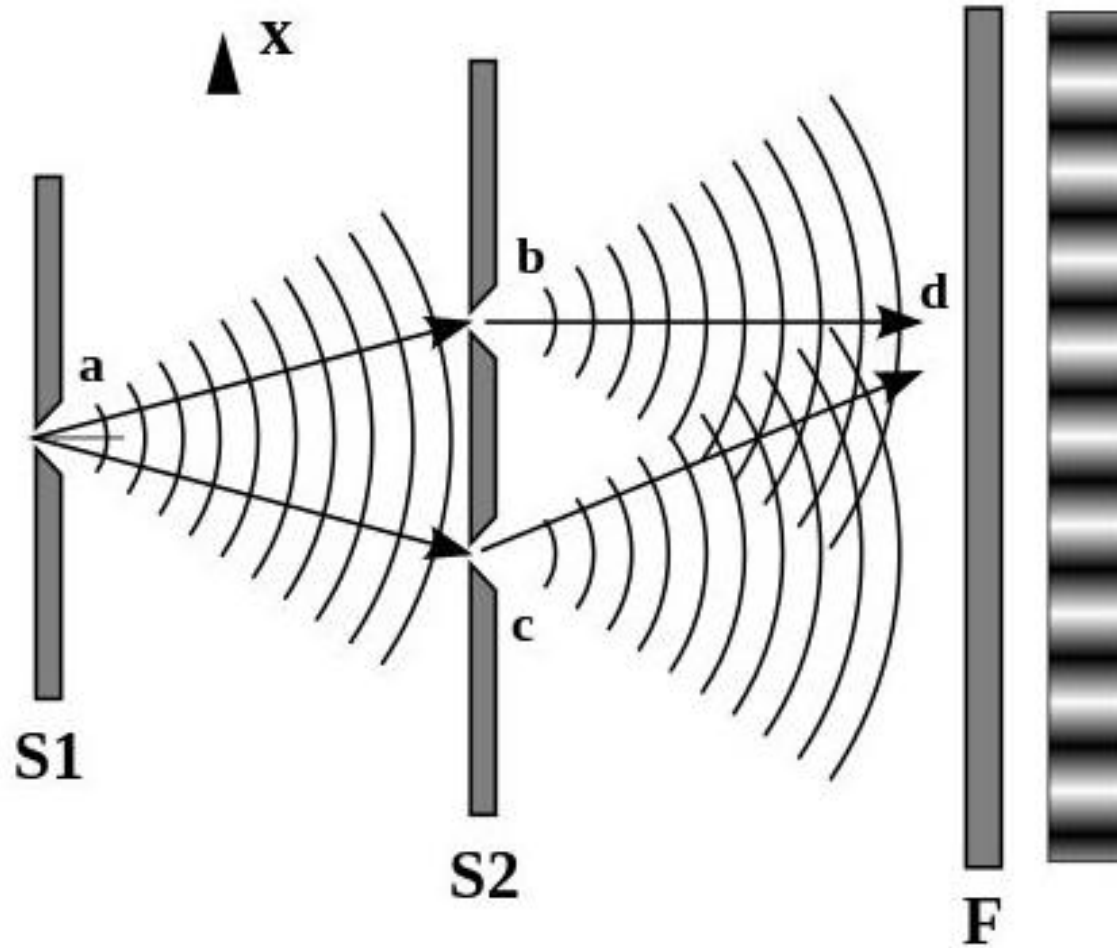
$$U \sim 0.1 \text{ V}, \lambda \sim 0.1 \text{ nm}$$

- Neutrons are mainly scattered by atomic nuclei, and light atoms also have strong scattering of neutrons, so they are often used to determine the position of hydrogen and carbon in the crystal.
- Neutrons have magnetic moments and are particularly suitable for studying the structure of magnetic substances.

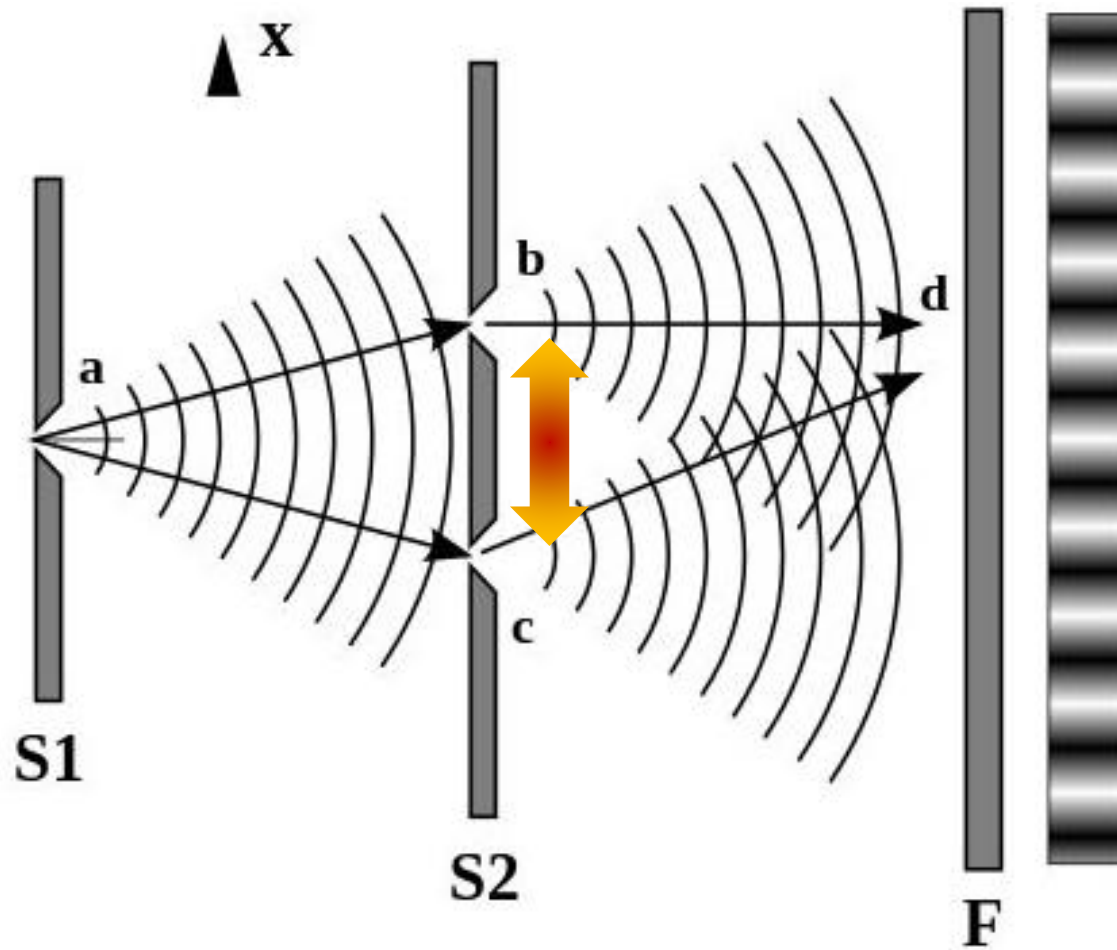
Homework

Assuming the velocity of the electrons is $1.5 \times 10^8 \text{ m s}^{-1}$ when reaching the metal target, what is the shortest wavelength for the X-ray and maximum energy for the photons

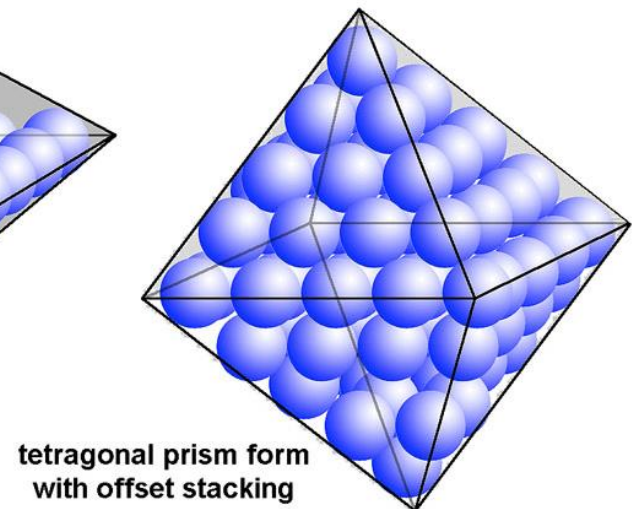
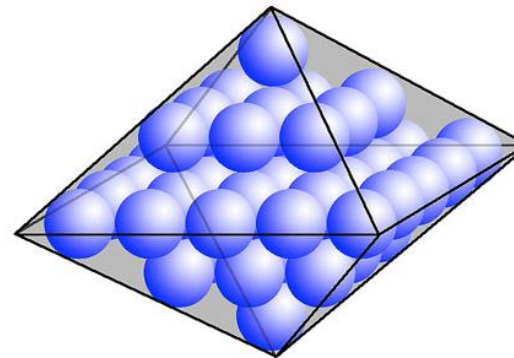
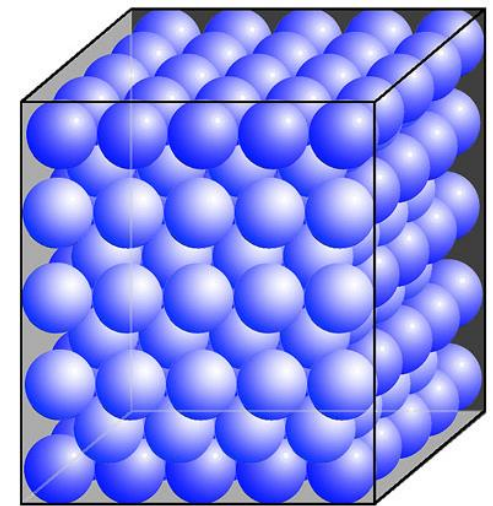
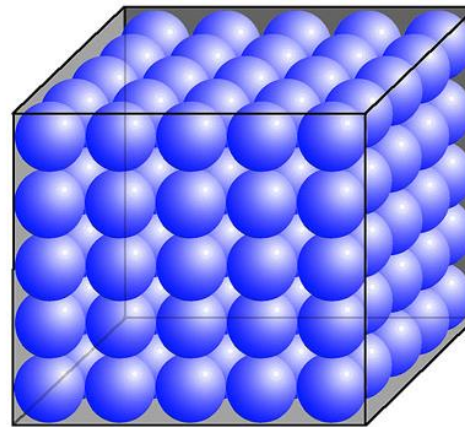
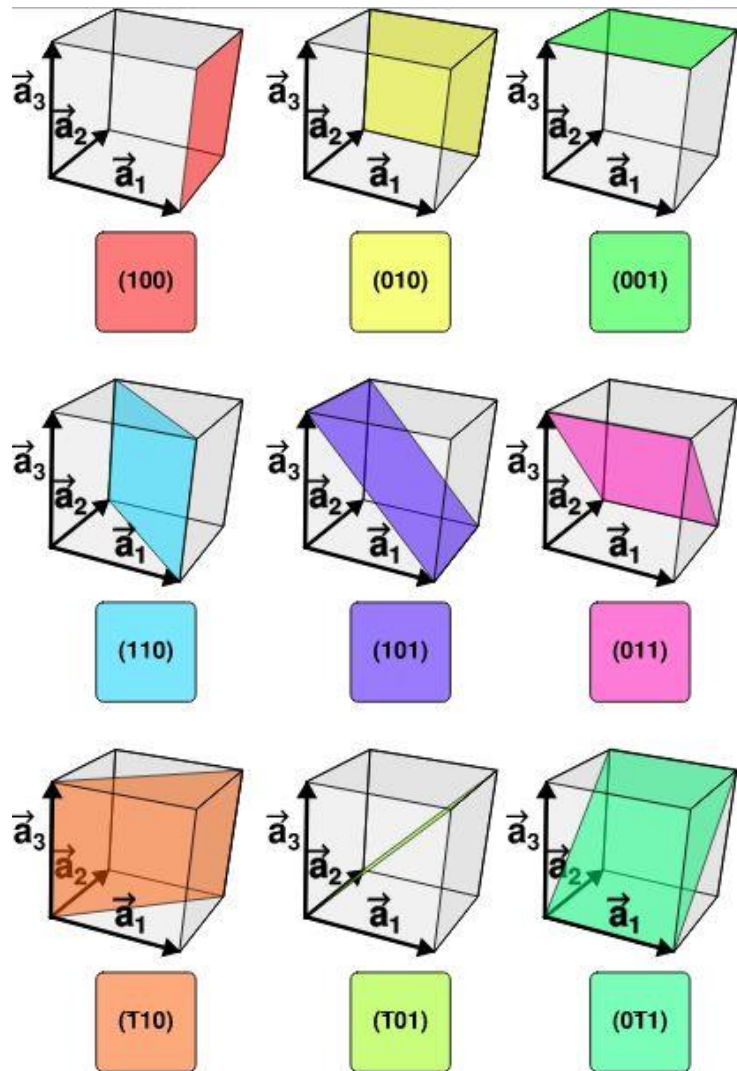
Optical grating and interference



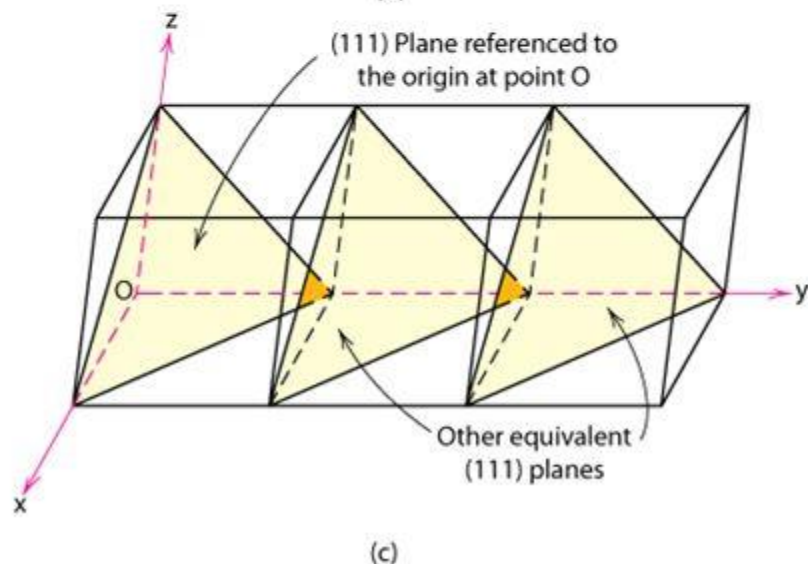
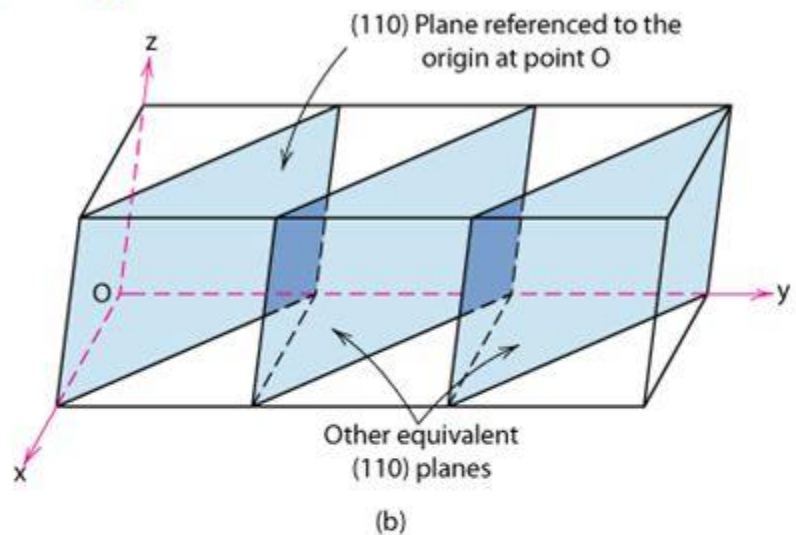
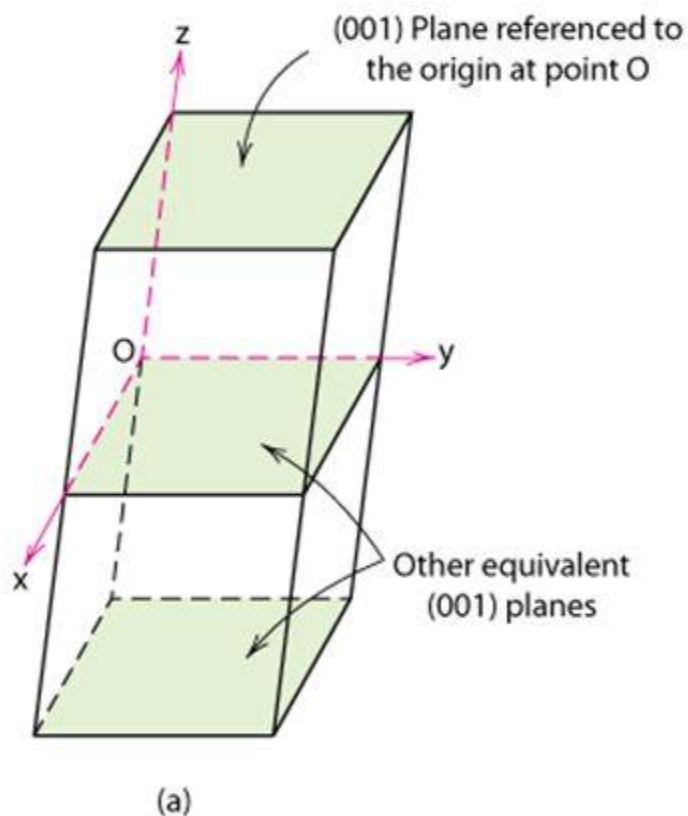
Optical grating and interference



Atom stacking and crystal planes

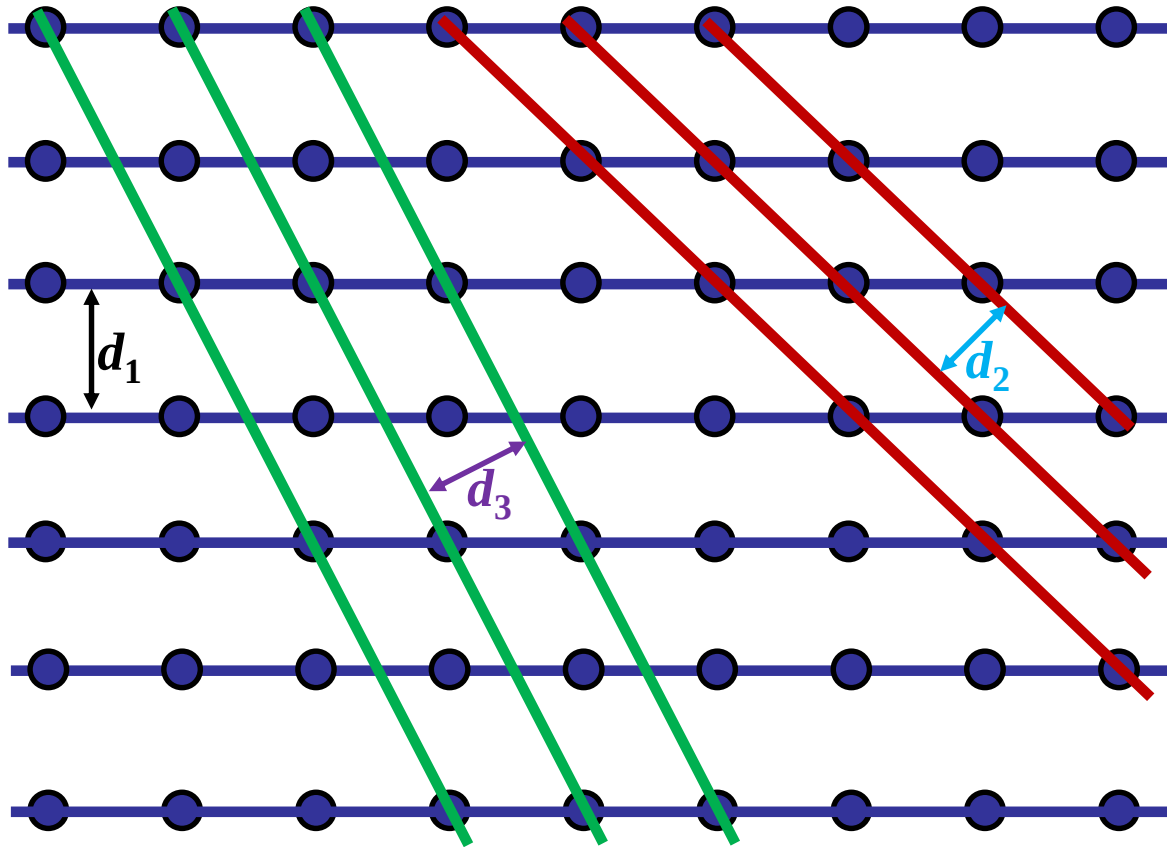


Crystallographic Planes

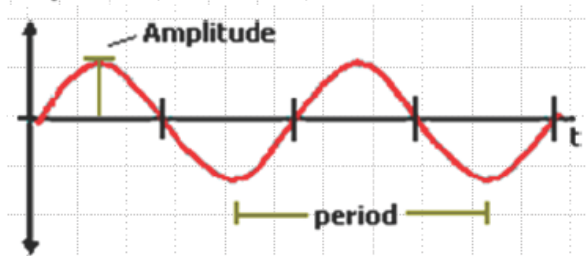
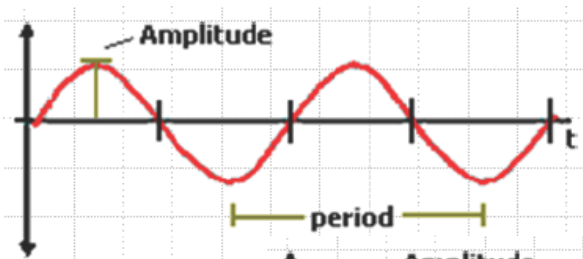
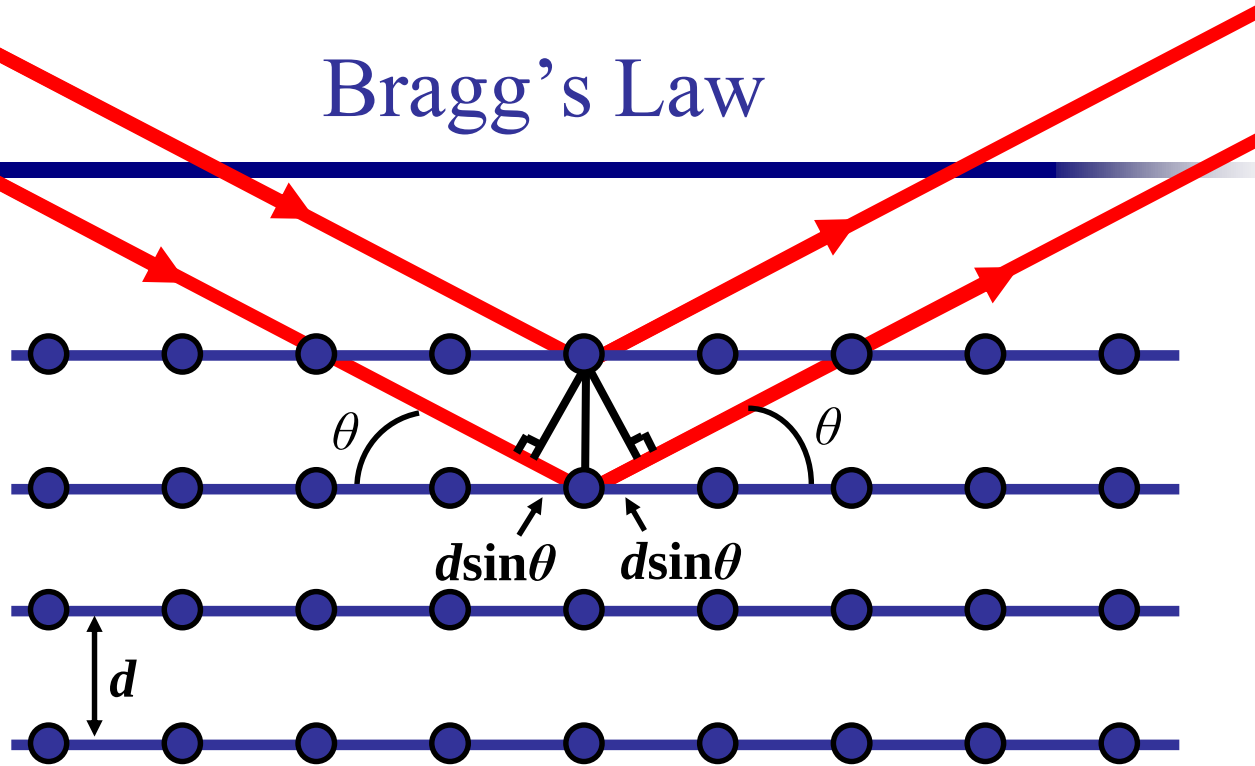


Adapted from Fig. 3.9, Callister 7e.

Bragg's Law



Bragg's Law

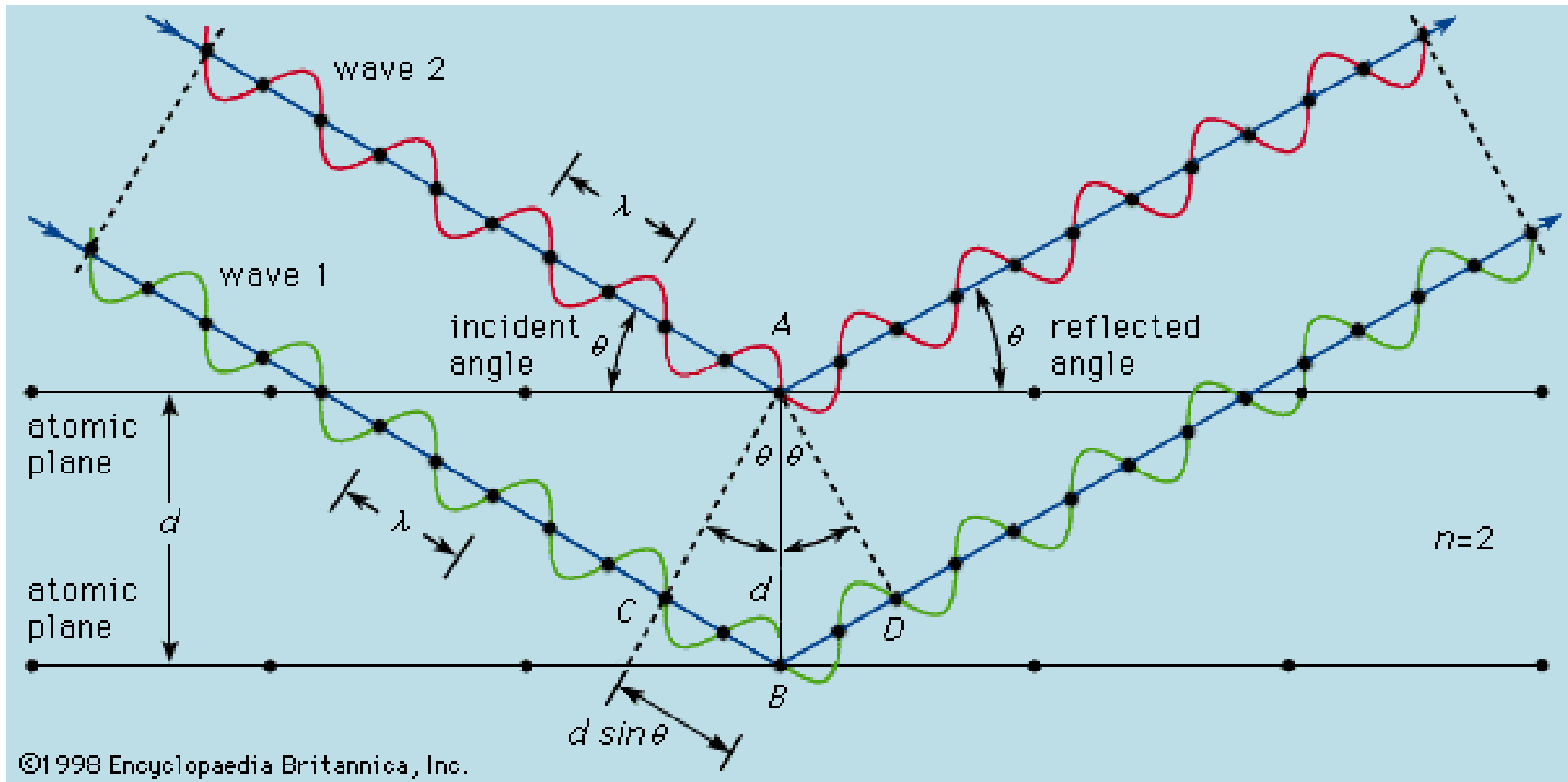


Path length difference = $2d \sin \theta$

Constructive interference for
 $2d \sin \theta = n \lambda$

Destructive interference for
 $2d \sin \theta = n \lambda / 2$

Bragg's Law

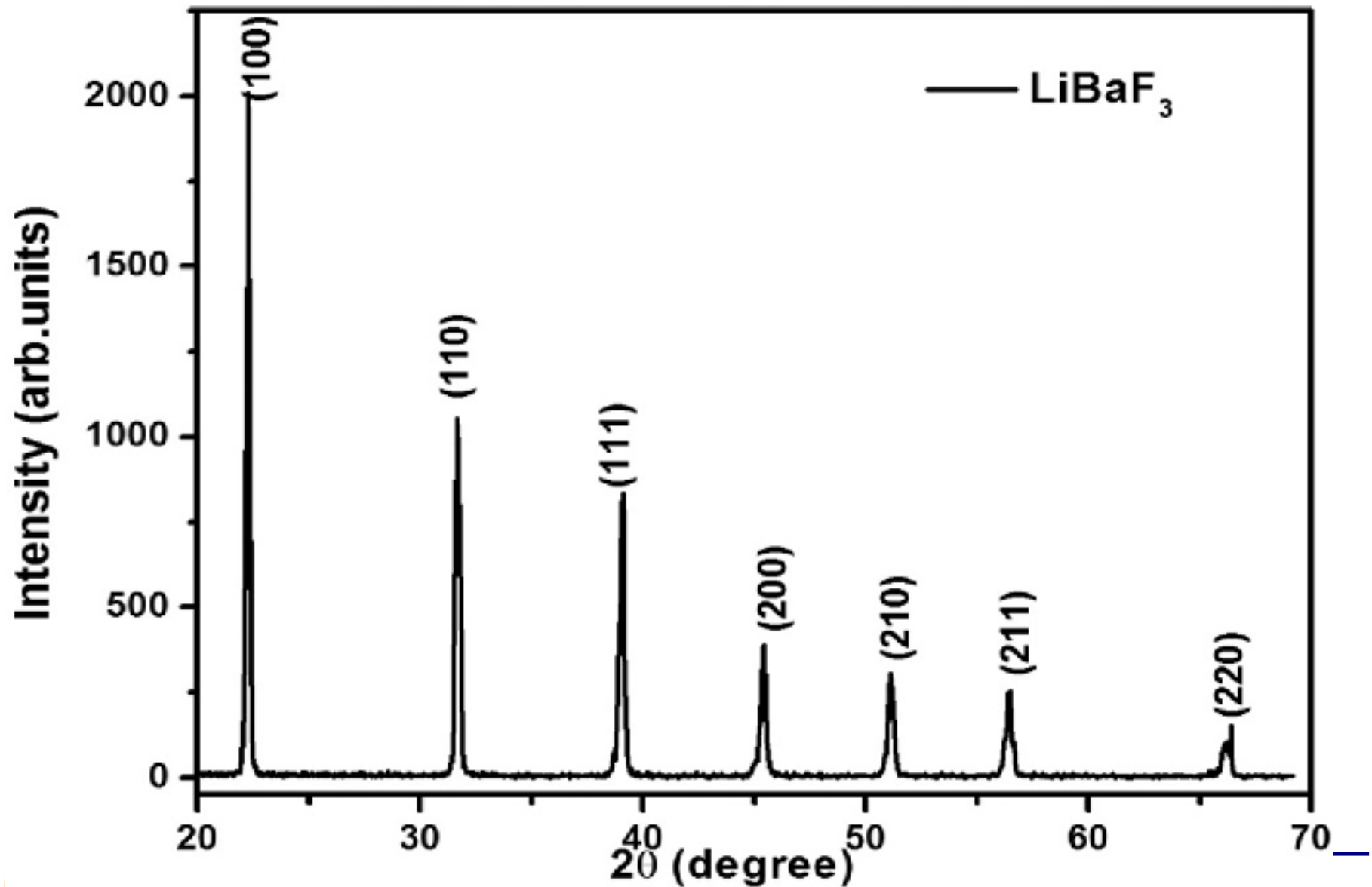


$$2d \sin \theta = n\lambda$$

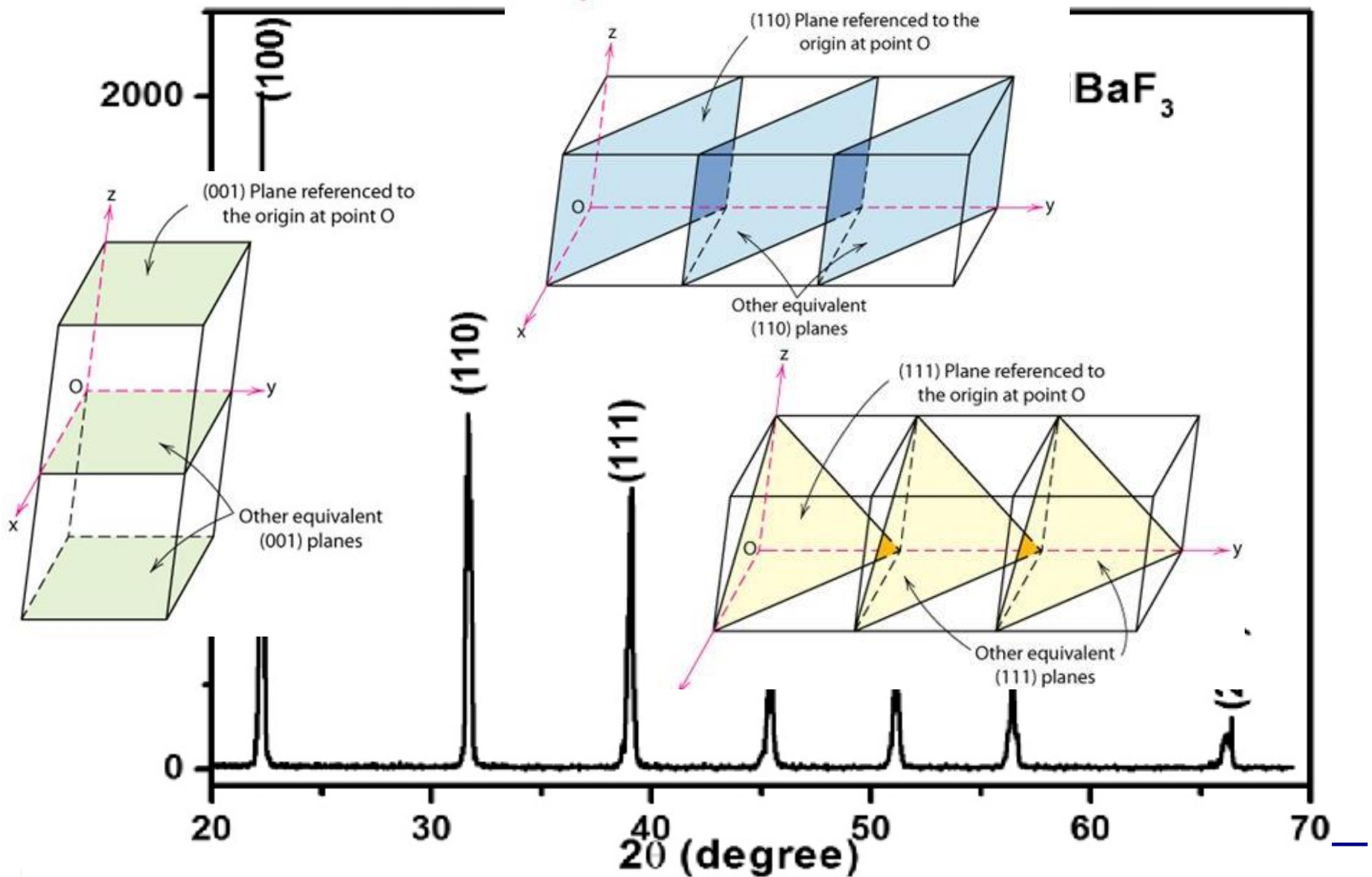
n : Order of reflection; d : Plane spacing; θ : Bragg Angle

Path difference must be integral multiples of the wavelength:

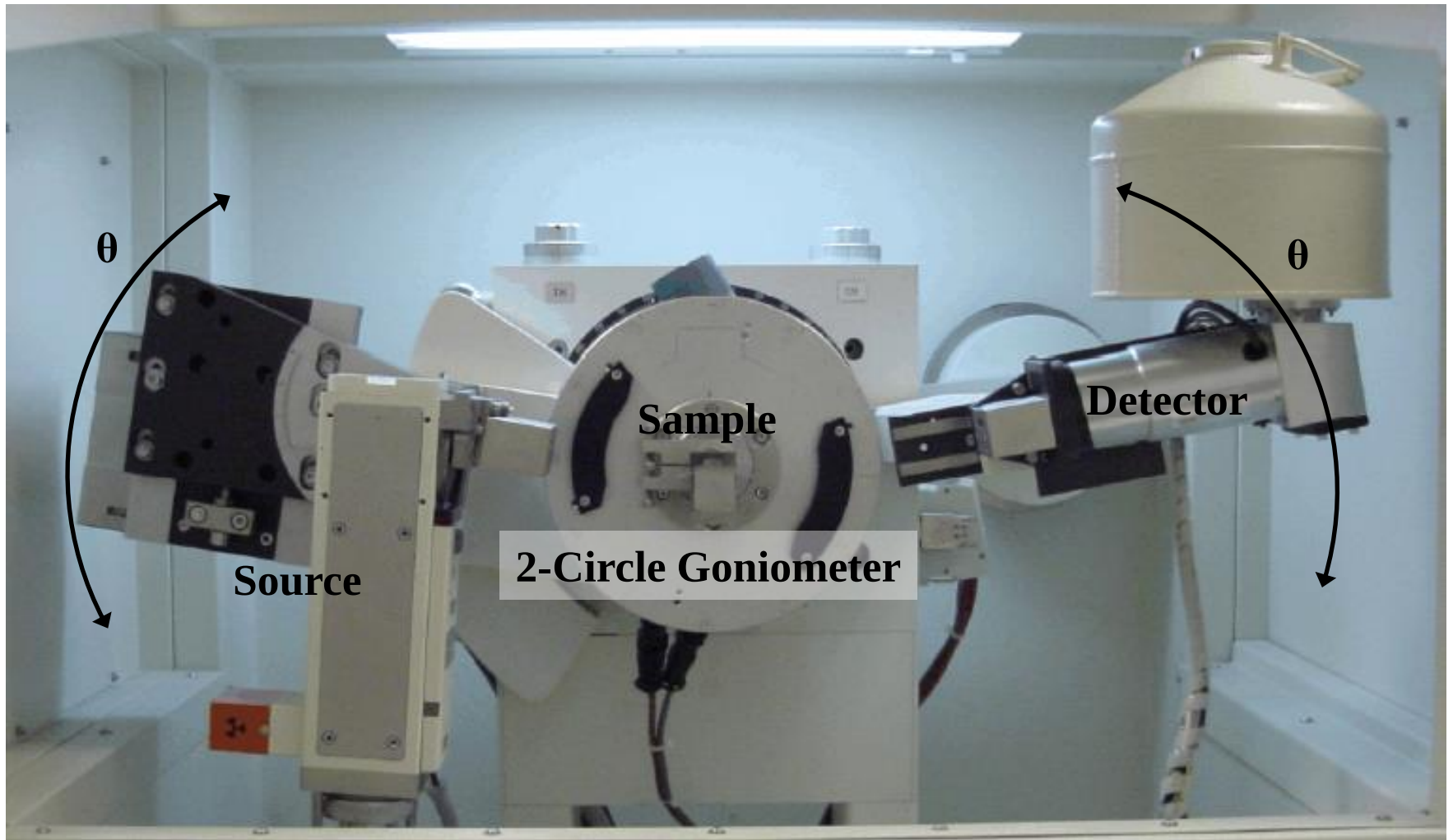
Typical Diffraction Spectra



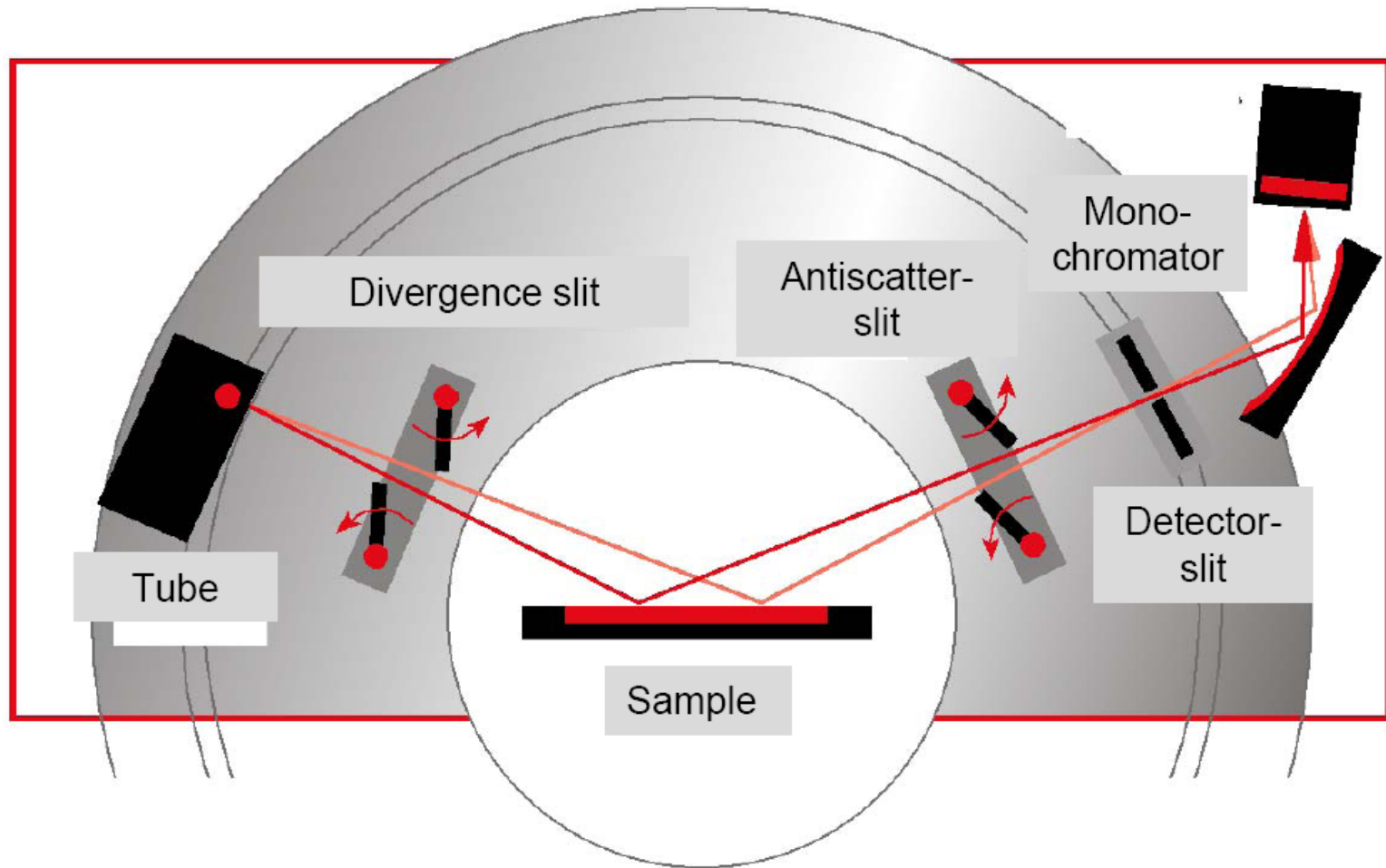
Typical Diffraction Spectra



Scintag Powder Diffractometer



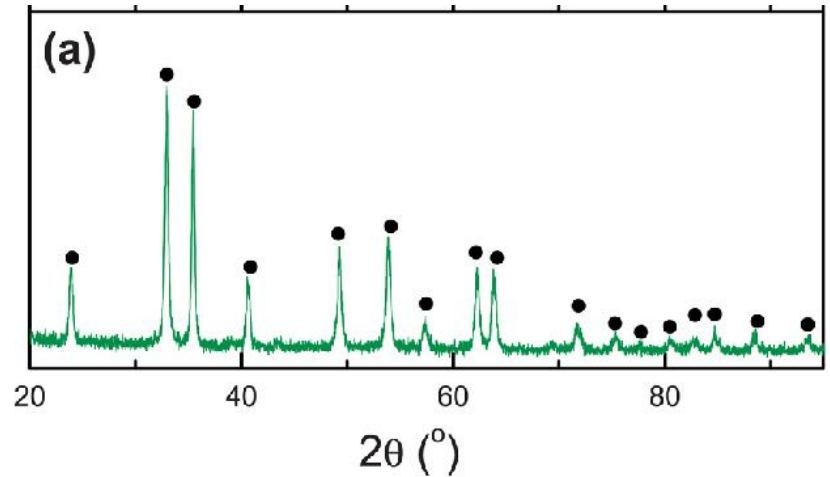
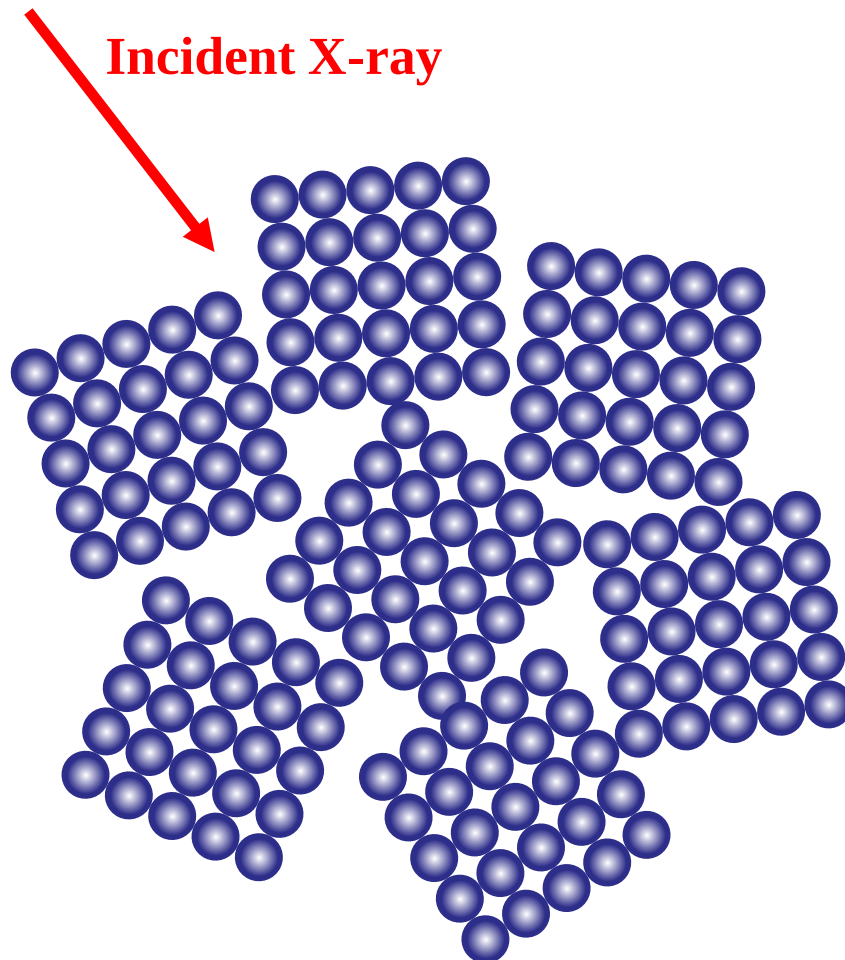
Typical Powder Diffraction System



- For flat samples

Powder Diffraction

Fine grained powder with random orientation



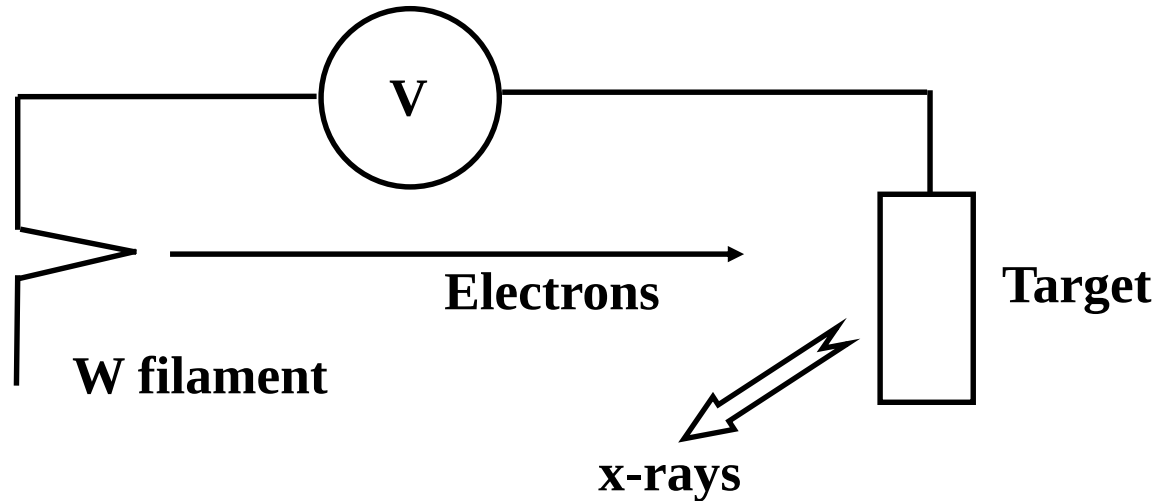
- All the crystal planes have been diffracted
- Grains have random orientation

Bruker D8 Discover Thin Film Diffractometer

Physics 216



X-Ray Sources



- Common target materials:
 - Mo
 - Cu
 - Co
 - Cr
 - Fe

} Produce x-rays of different wavelength
- Only ~1% of total energy goes to creating x-rays.

X-Ray Spectrum

1. Short wavelength limit

- All kinetic energy of electron transferred to photon.

$$eV = h\nu_{\max} = \frac{hc}{\lambda_{\text{swl}}}$$

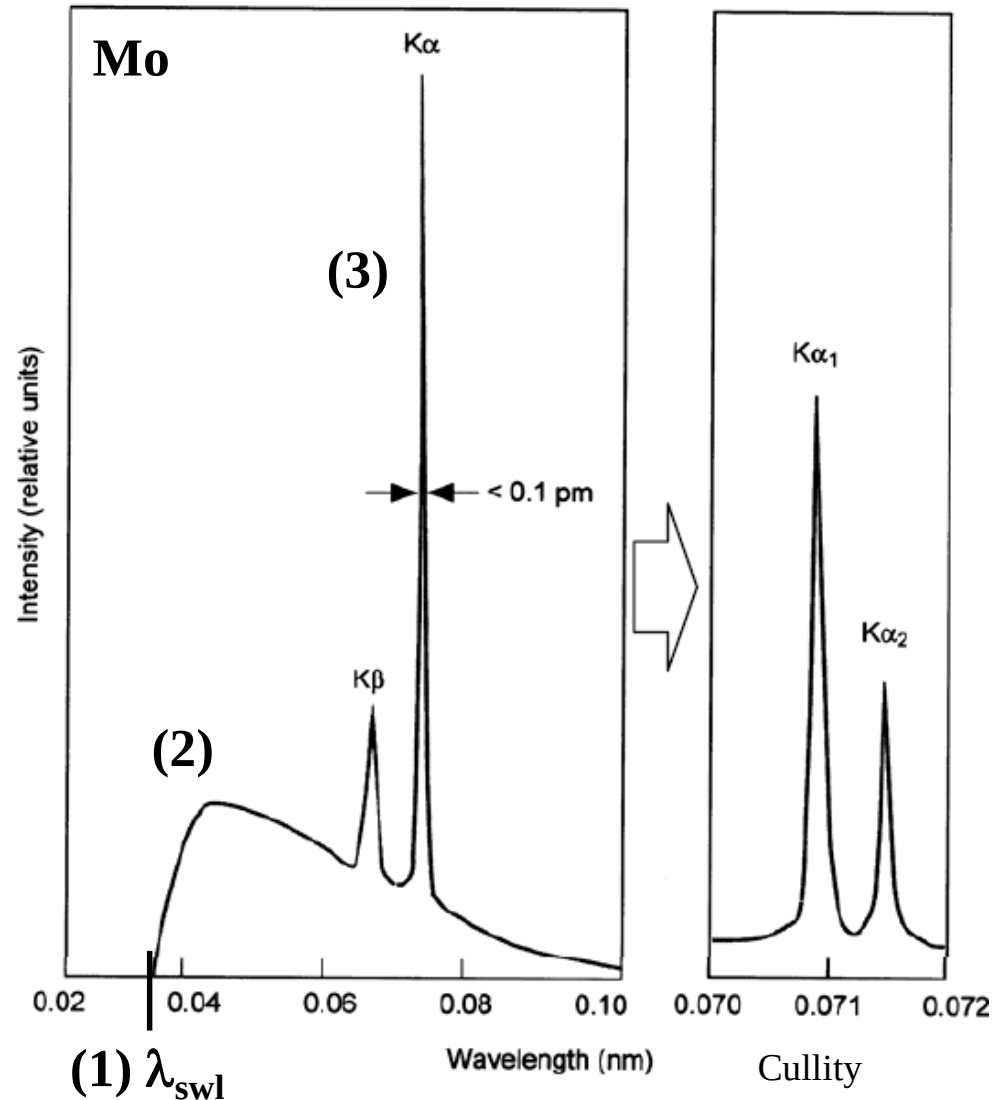
$$\lambda_{\text{swl}} = \frac{12.398 \times 10^3}{V} \quad \lambda \text{ in } \text{\AA} \quad V \text{ in volts}$$

2. Bremsstrahlung radiation or “braking radiation”

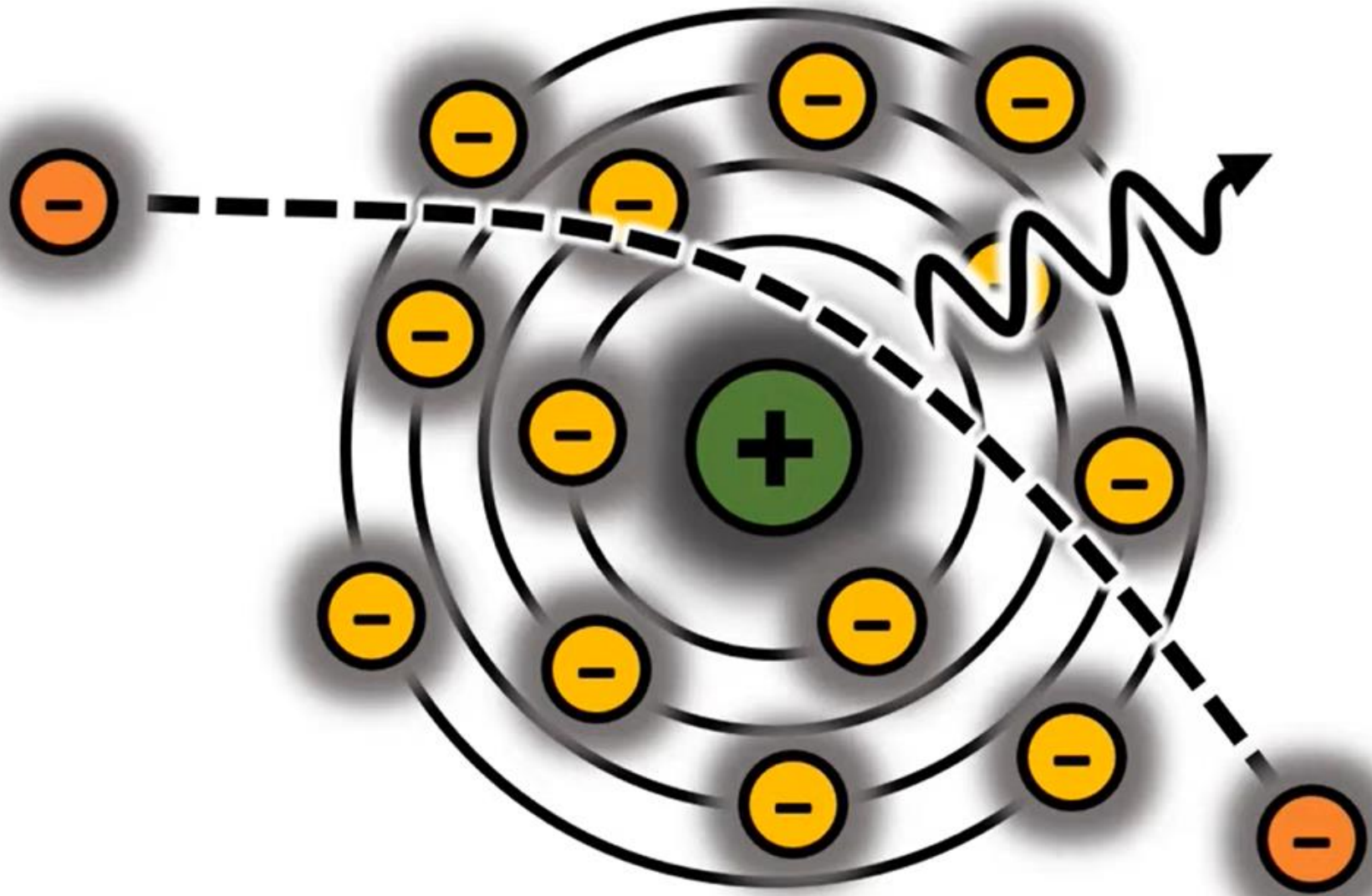
- Deceleration of electrons by target atoms results in radiation of x-rays.

3. Characteristic radiation

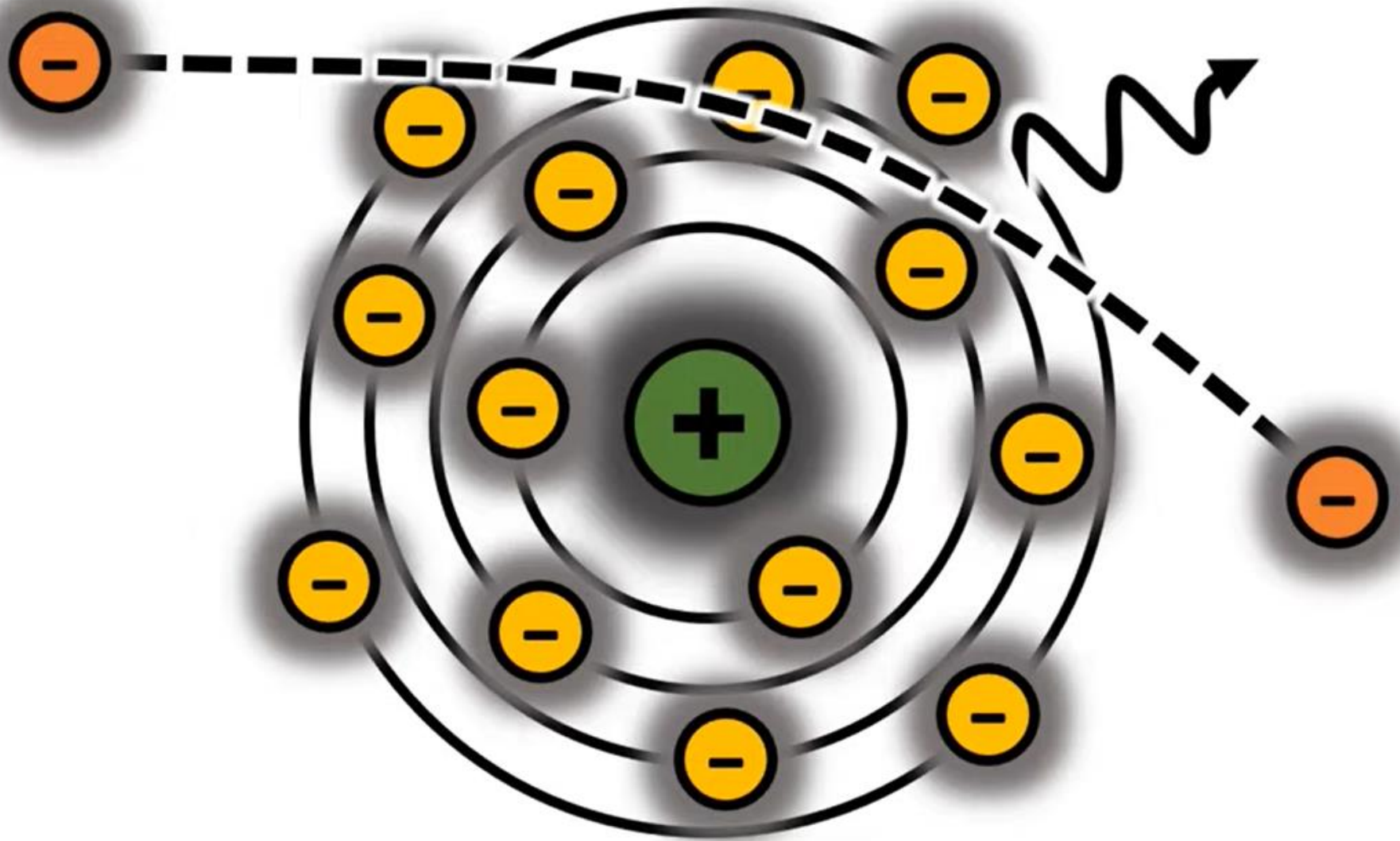
- Discrete intra-atomic electron transitions within target atoms.



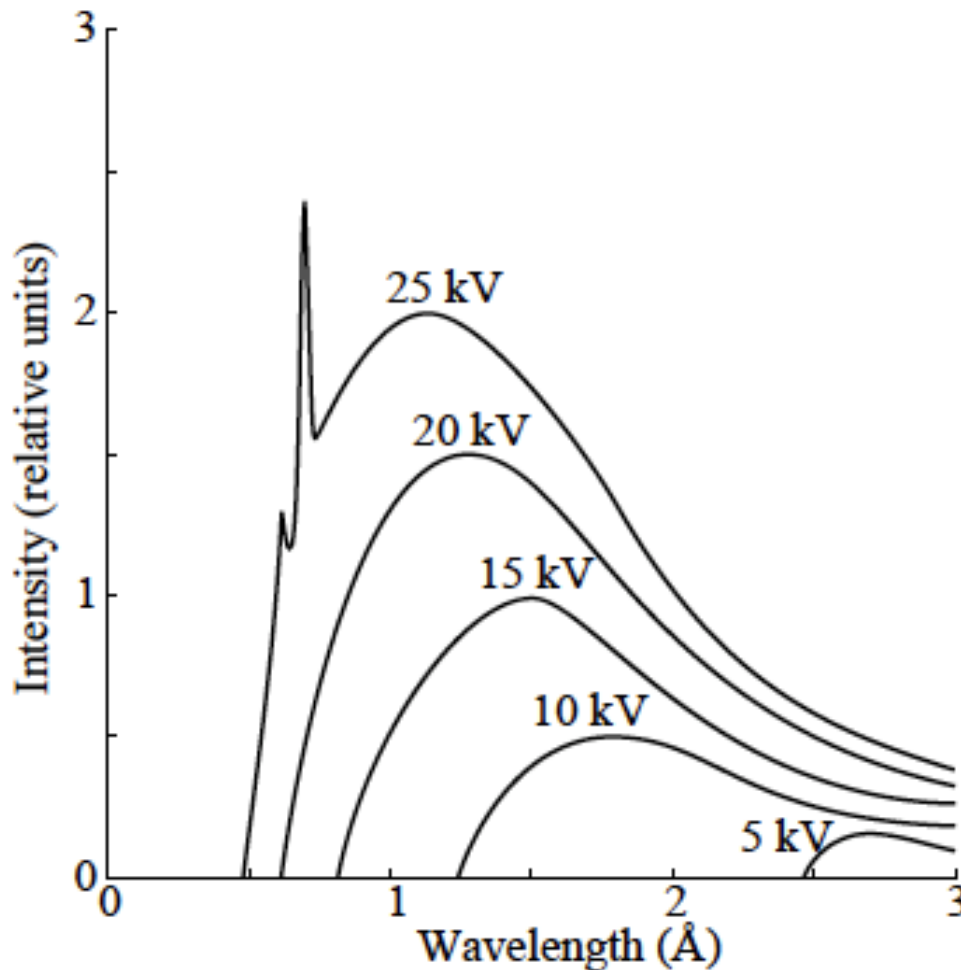
Bremsstrahlung radiation: close to nuclei



Bremsstrahlung radiation: far away from nuclei



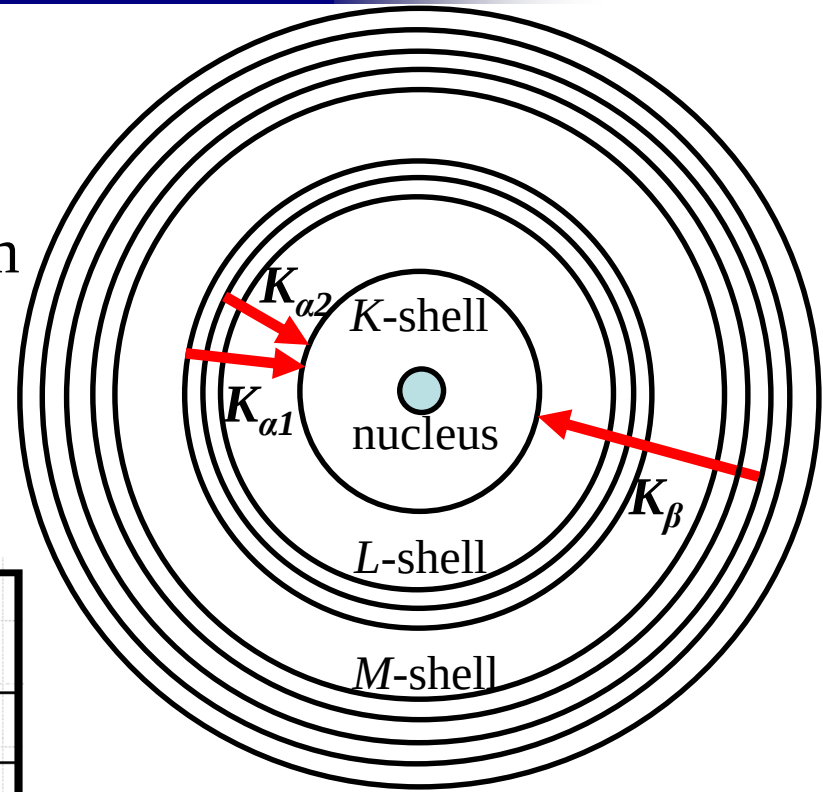
Characteristic radiation of X-ray



Characteristic radiation of X-ray as acceleration voltage is high enough to knock out the electrons of the targeted element

Characteristic Radiation

1. Incoming electron knocks out a K -shell electron.
2. Vacancy is filled by electron from L -shell (K_{α} radiation) or M -shell (K_{β} radiation) resulting in emission of radiation



Some Commonly Used X-ray K wavelengths				
Cullity	(Å)			
Element	K_{α} (av.)	$K_{\alpha 1}$	$K_{\alpha 2}$	$K_{\beta 1}$
Cr	2.29100	2.28970	2.29361	2.08487
Fe	1.93736	1.93604	1.93998	1.75661
Co	1.79026	1.78897	1.79285	1.62079
Cu	1.54184	1.54056	1.54439	1.39222
Mo	0.71073	0.70930	0.71359	0.63229

$$I_{K-line} = Bi(V - V_K)^n$$

B=constant

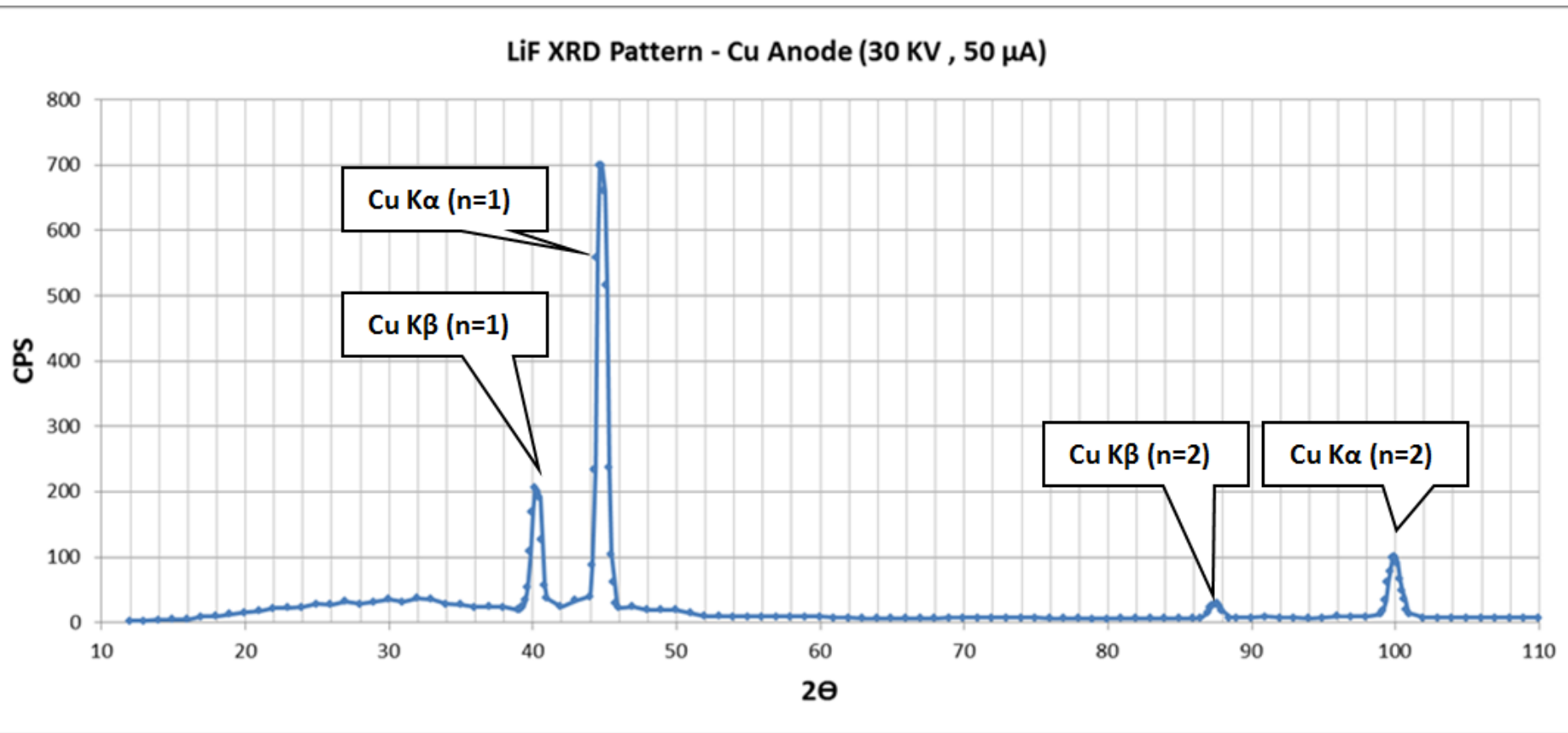
i=tube current

V=tube voltage

V_K =K critical excitation voltage

$n \approx 1.5$

K_α and K_β radiation for the same crystal plane



- For the same crystal plane, two or more diffraction peaks appear
- A filter has to be used to block the unwanted x-ray

X-Ray Absorption

$$I = I_0 e^{-\mu_l x}$$

$$\mu_l = \mu_m \rho = (\sigma_m + \tau) \rho$$

I_0 = initial intensity

μ_l = linear absorption coefficient

μ_m = mass absorption coefficient

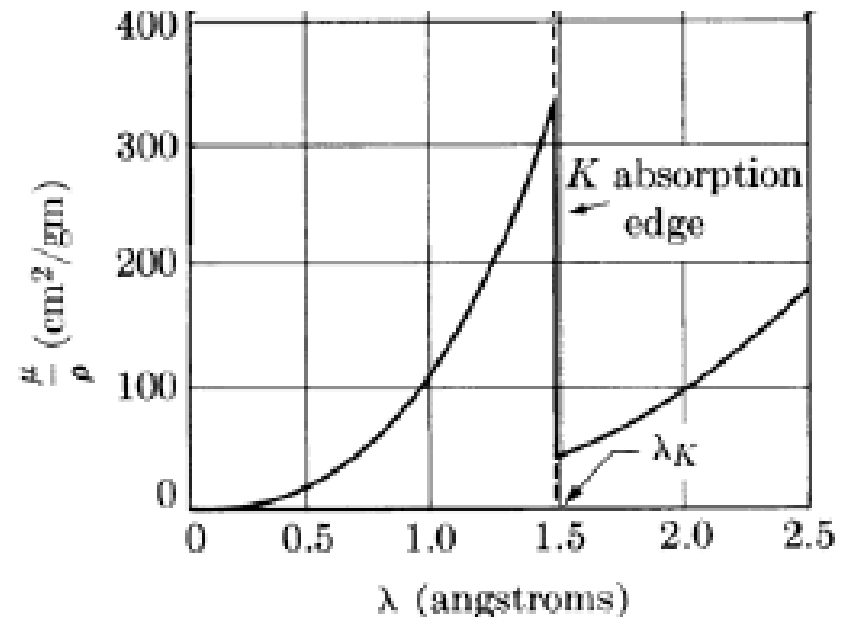
τ = true absorption coefficient

ρ = density

- Mass absorption coefficient increases according to:

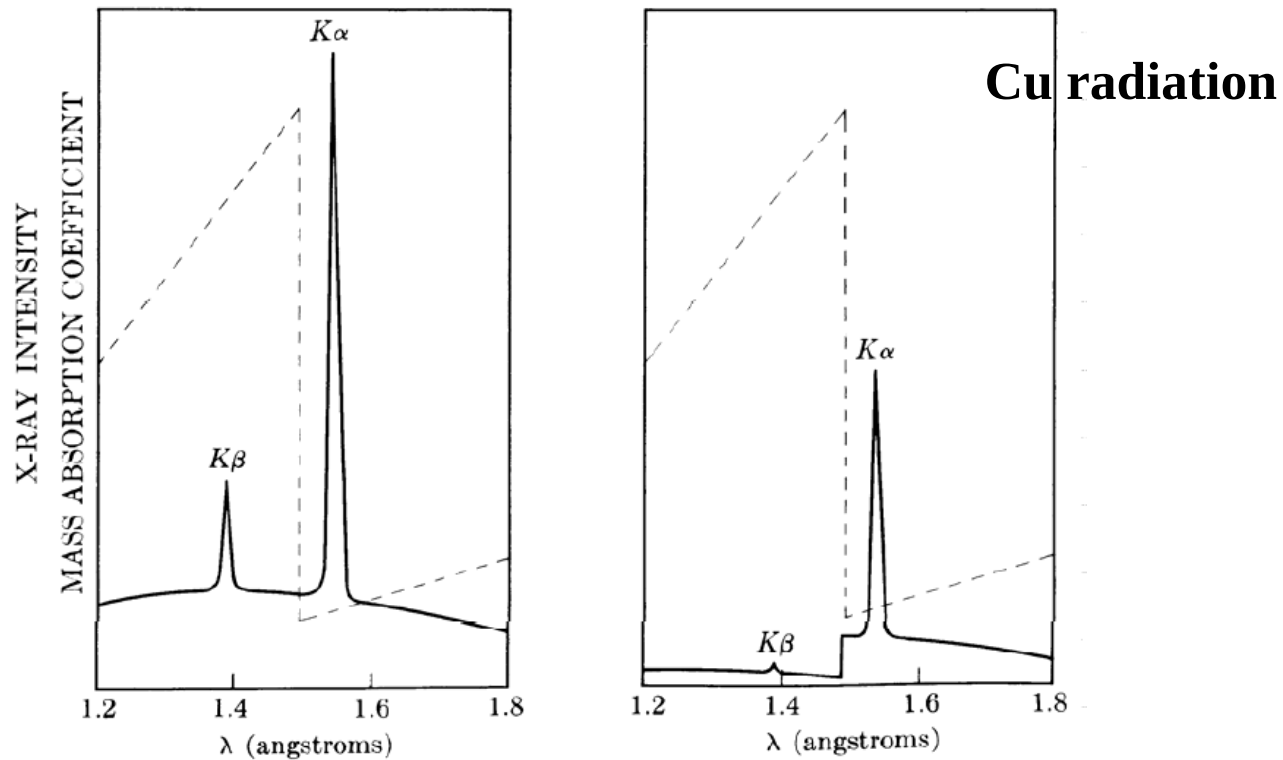
$$\mu_m = C \lambda^3 Z^3$$

- Sharp drop at absorption edges:
 - energy (wavelength) that causes ejection of a K shell electron



Use of Filters

- X-ray diffraction requires monochromatic radiation.
- Use a filter/absorber with an absorption edge just above the undesirable radiation.



No Filter

Ni Filter

Cullity

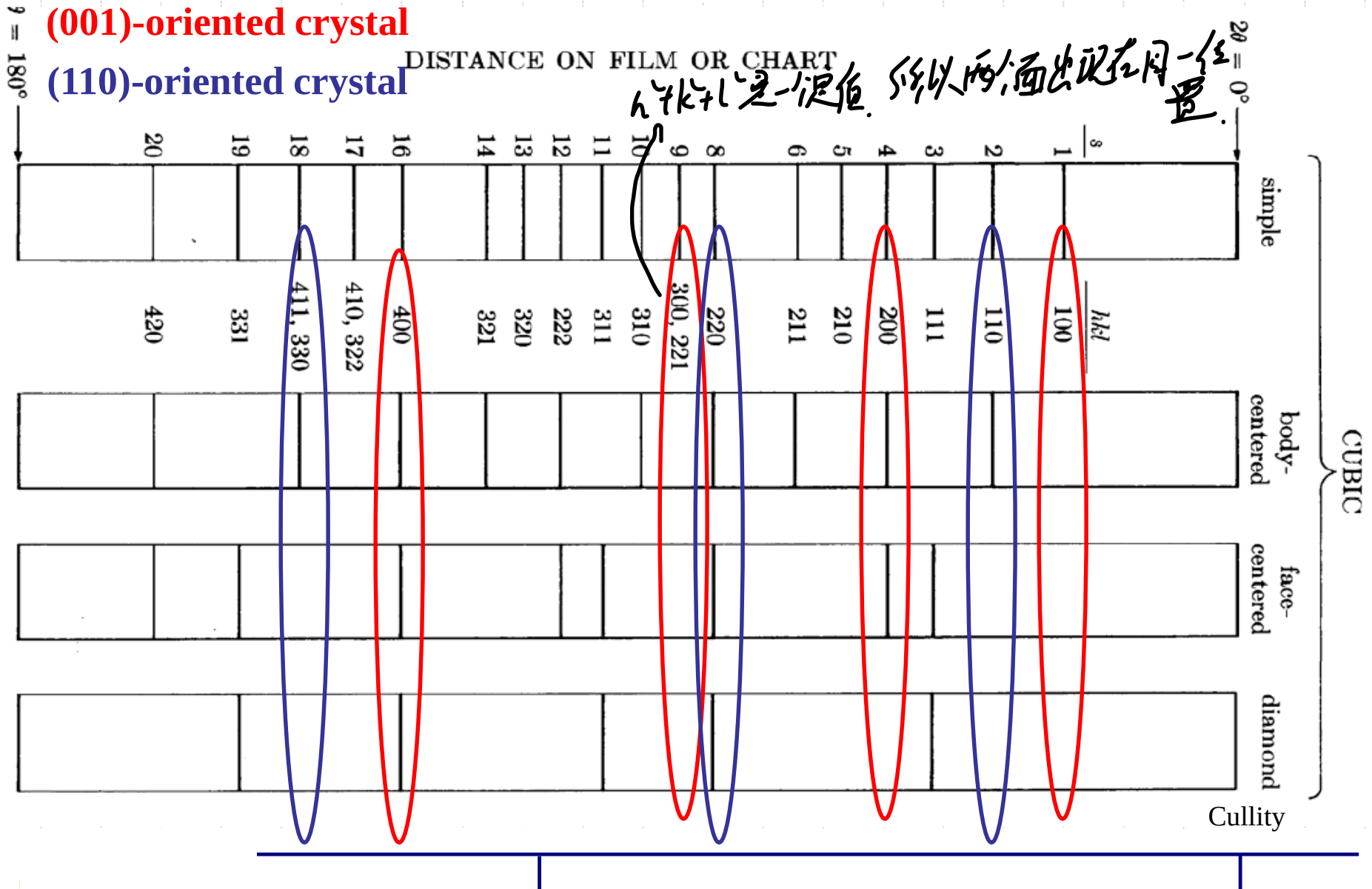
Types of Filters

Filters for Suppression of $K\beta$ Radiation

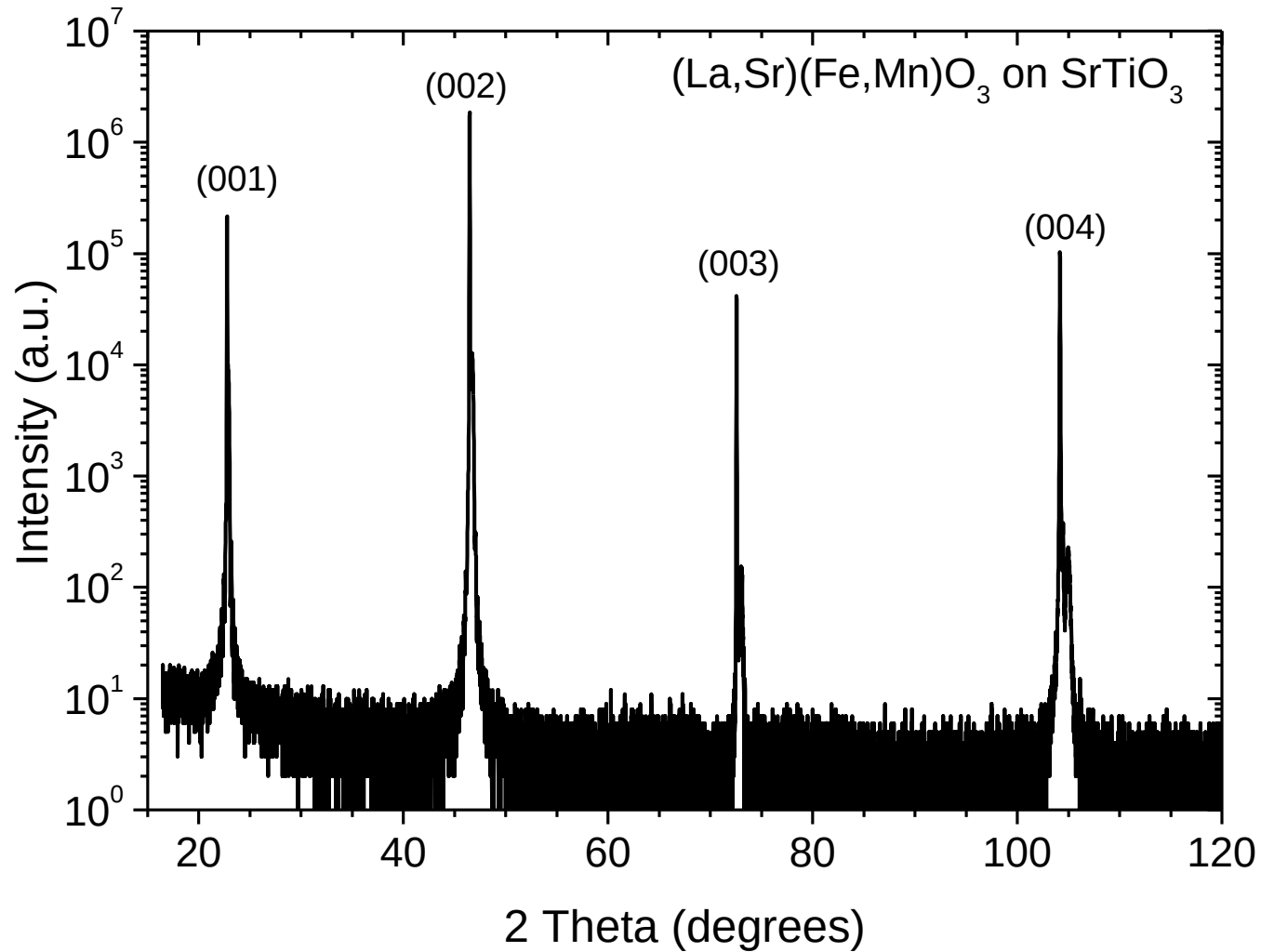
Target	Filter	Incident beam $\frac{I(K\alpha)}{I(K\beta)}$	Filter thickness for $\frac{I(K\alpha)}{I(K\beta)} = \frac{500}{1}$ in trans. beam		$\frac{I(K\alpha) \text{ trans.}}{I(K\alpha) \text{ incident}}$
			mg/cm ²	in.	
Mo	Zr	5.4	77	0.0046	0.29
Cu	Ni	7.5	18	0.0008	0.42
Co	Fe	9.4	14	0.0007	0.46
Fe	Mn	9.0	12	0.0007	0.48
Cr	V	8.5	10	0.0006	0.49

Thin Film Diffraction

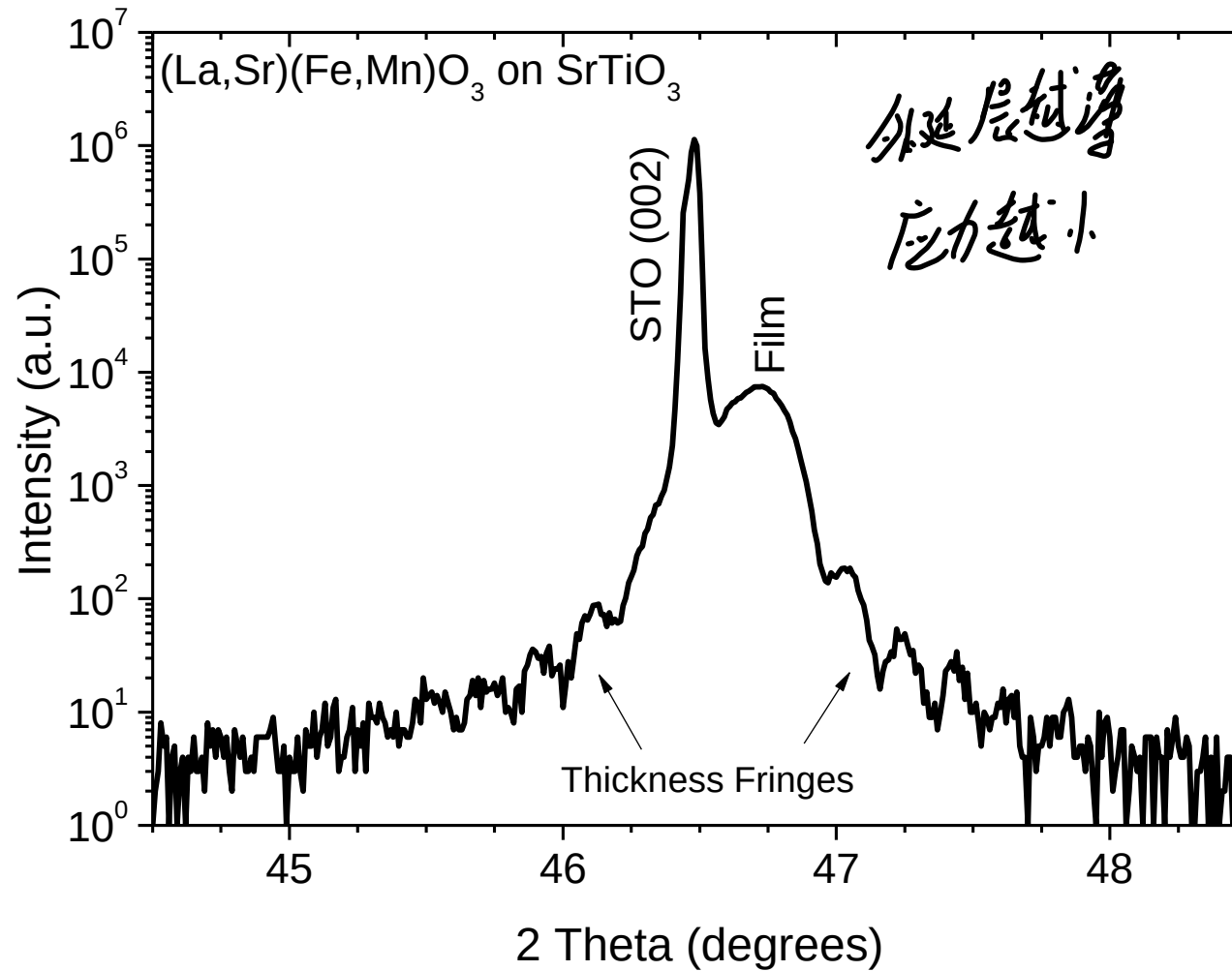
Cubic Crystals



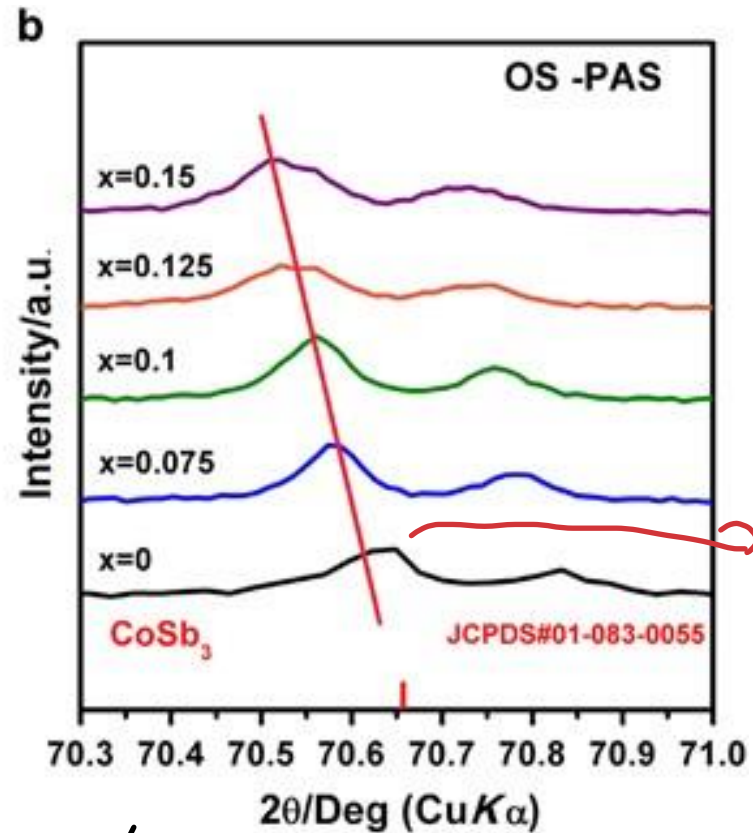
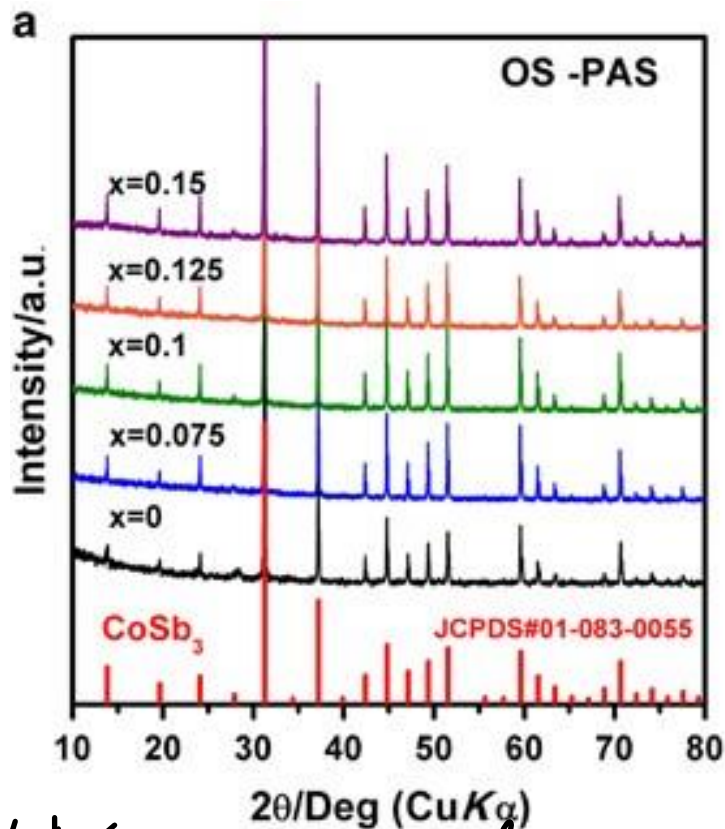
Thin Film Diffraction Pattern



Thin Film Diffraction Pattern



Peak shifts



掺杂浓度 ↓ → $d \downarrow \rightarrow \Delta \uparrow$

$$2d\sin\theta = n\lambda$$

d is associated with lattice constant, a

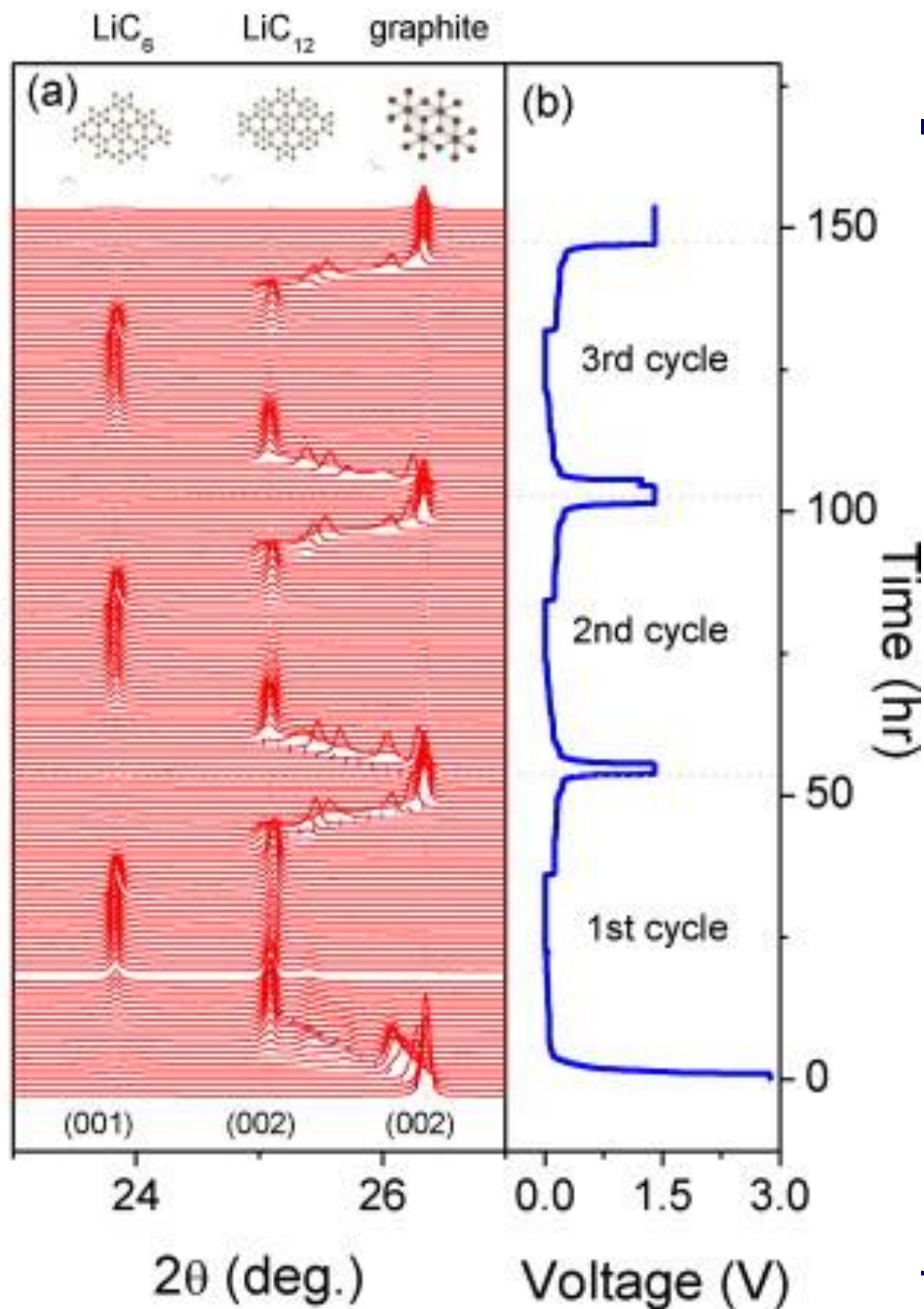
掺杂使晶格常数发生变化

杂物

Peak shift is due to variation of the lattice constant

➤ Strain, doping, etc

Intensity (arb. unit)

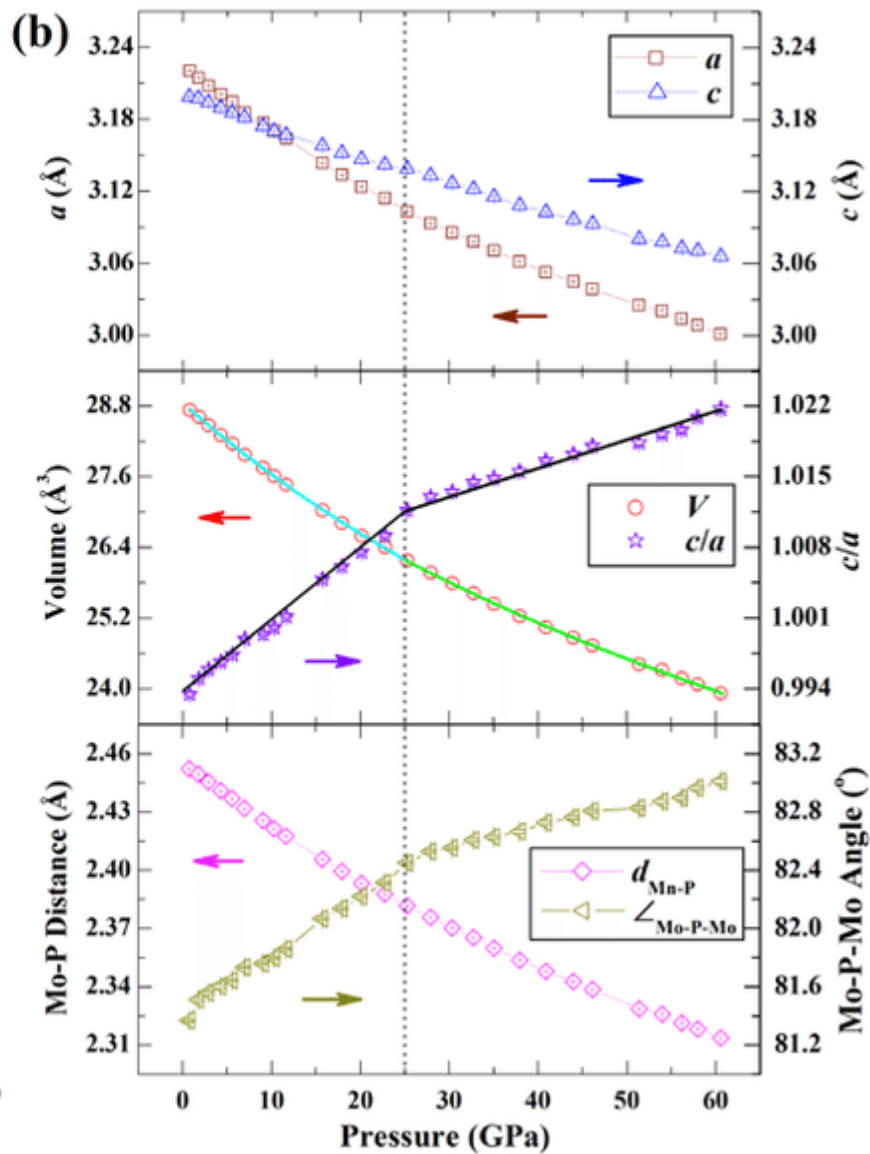
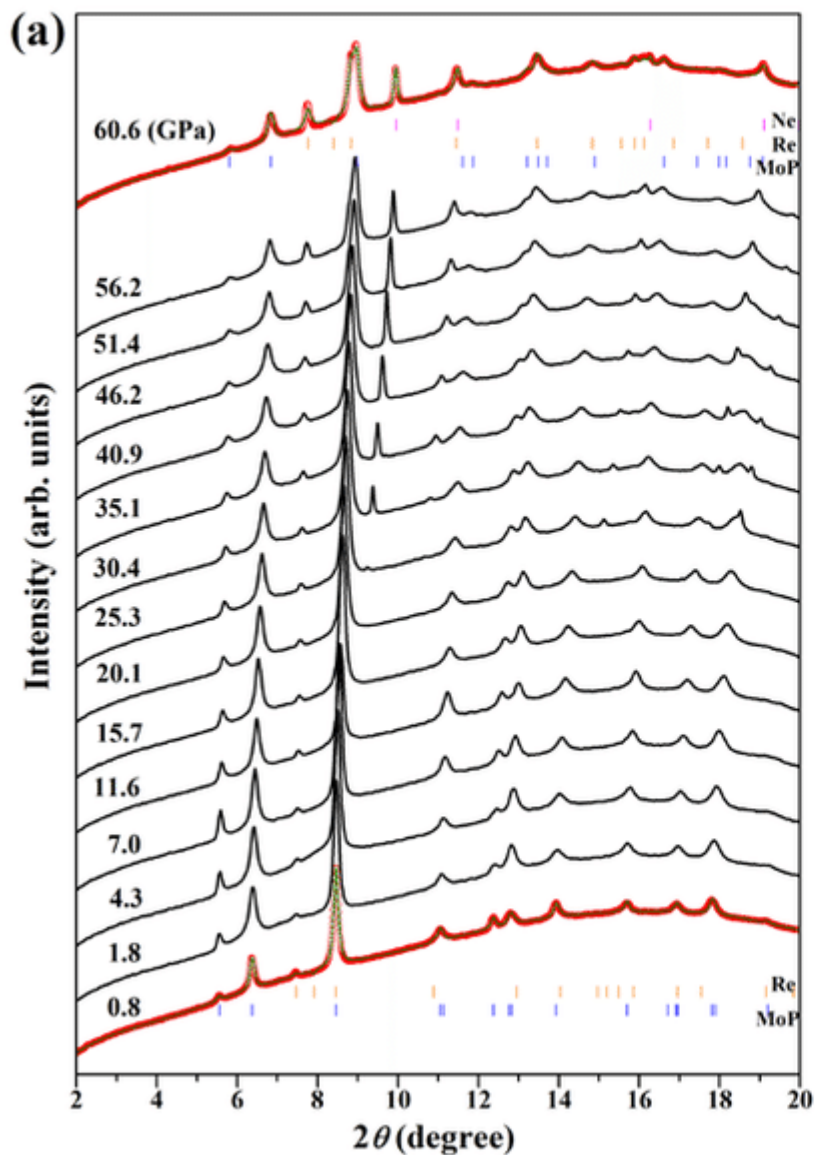


原位监测

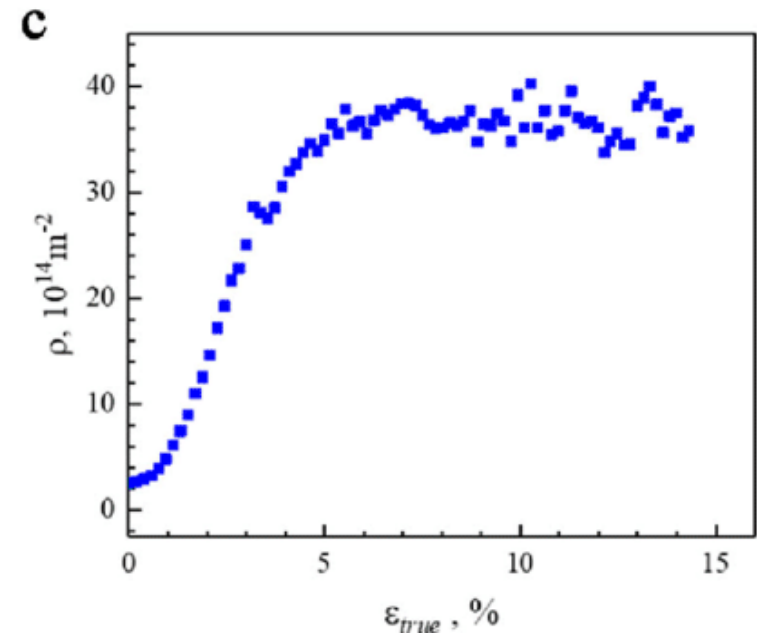
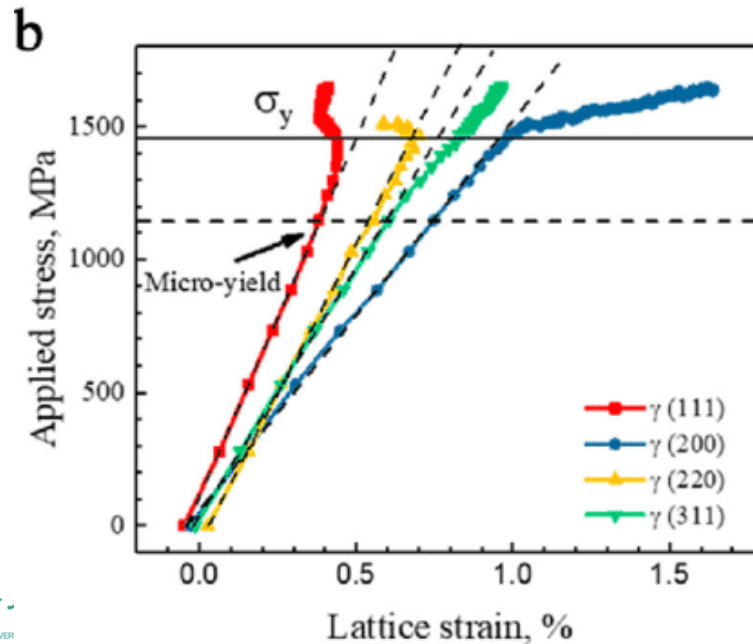
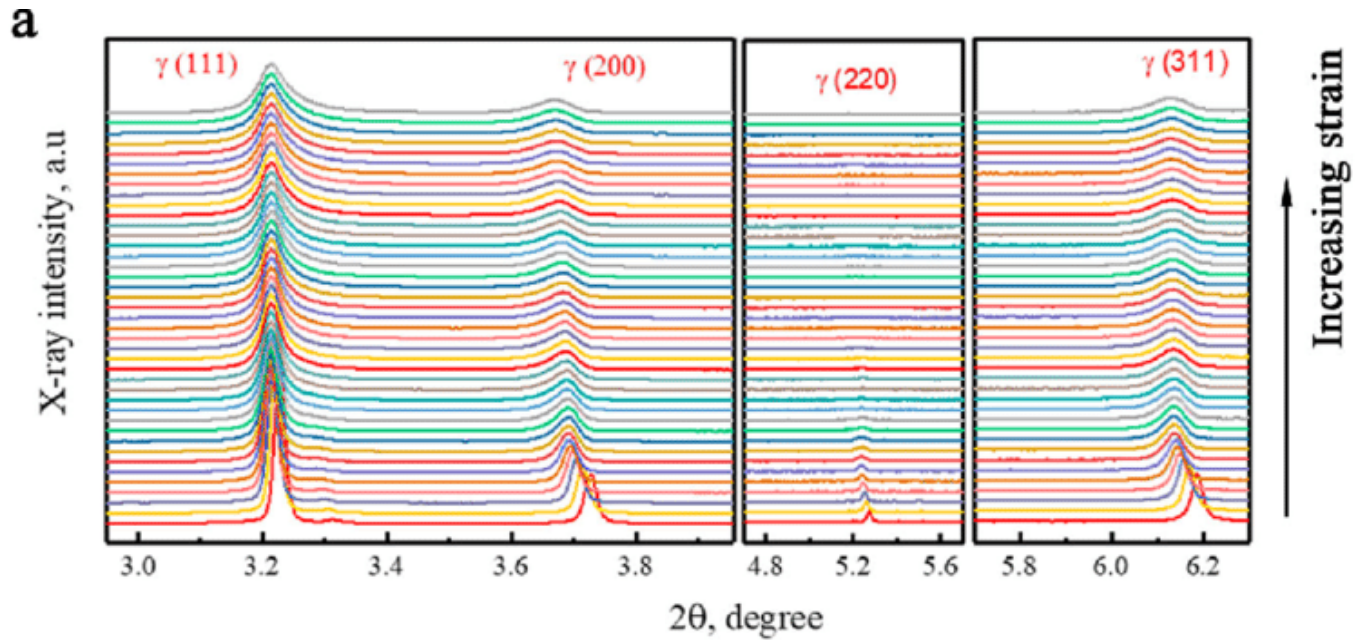
In situ XRD under
ion intercalation

在特定样品如电池电极的工作
状态下进行监测。

In situ XRD at various pressure



In situ XRD at varied strain



Summary

- X-ray Diffraction is a very useful to characterize materials for following information
 - Phase analysis
 - Lattice parameter determination
 - Strain determination
 - Texture and orientation analysis
 - Order-disorder transformation
 - and many more things
- Choice of correct type of method is critical for the kind of work one intends to do.
- Powerful technique for thin film characterization